

10. Universal Transverse Mercator (UTM)

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Arc Unit : Degree

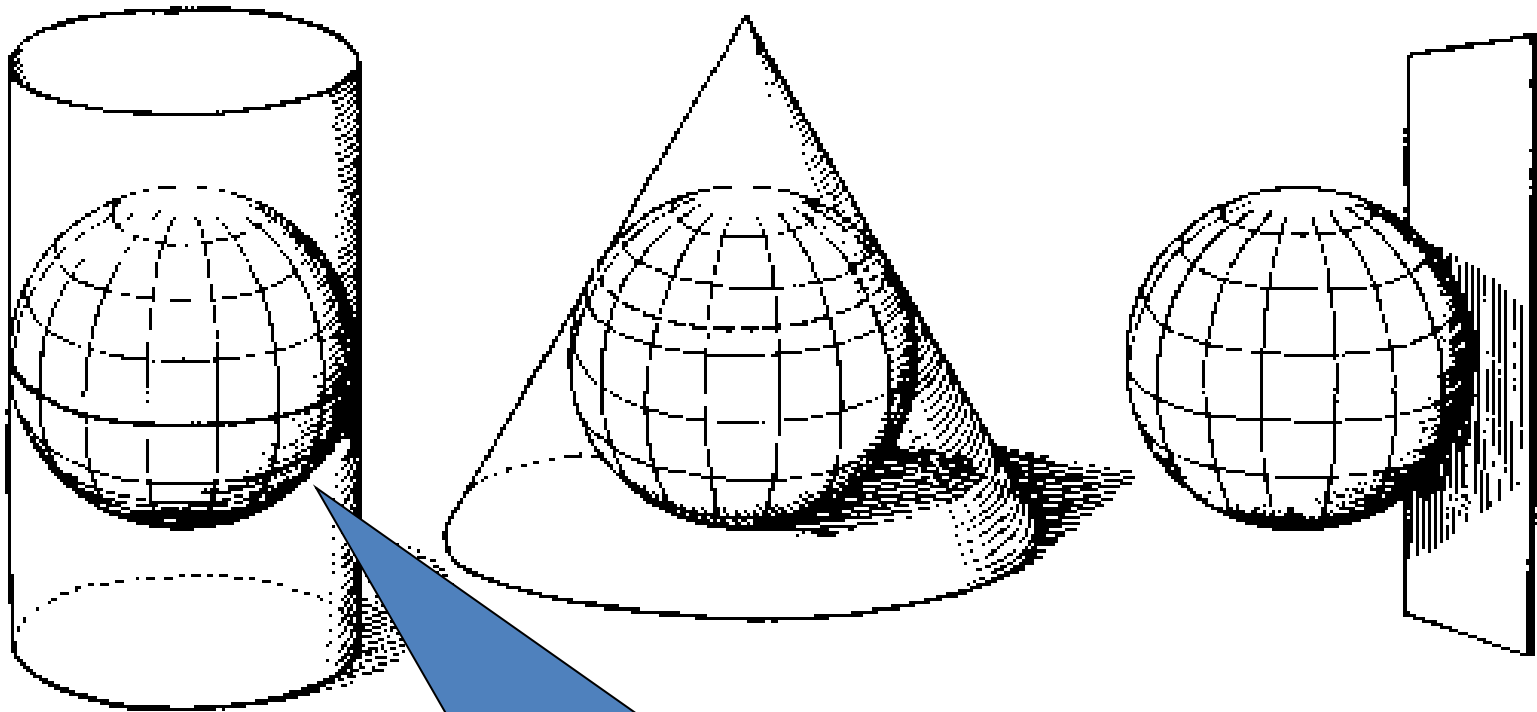
- DMS : sexagesimal Notation
 - Use of degrees-minutes-seconds is also called DMS notation. , arcminute, arcsecond
 - E.g. $45^{\circ} 50' 55''$ | 45:50:55 | 45d50m55.12345s
 - $45^{\circ} 50.124'$
 - $45^{\circ} 50' 55.12345''$ (mm-5 digits equiv. mm)
- DD : Decimal Degree Notation
 - E.g. 45.123456789 (mm- 9 digits equiv. mm)
- Maritime charts are marked in degrees and decimal minutes to facilitate measurement
 - 1 minute of latitude is 1 nautical mile.
 - E.g. $45^{\circ} 50.91667'$ or 45d 50.91667m

Linear (meter) vs. Angular (degree) Measurement

- $R_{\text{Earth}} = 6,371 \text{ km.}$
- 1 degree $\sim$ 111,000 meter
 - **1 mm $\sim$ 0.000 000 001 degree (9-digits)**
- 1 minute $\sim$ 1,800 meter
- 1 second $\sim$ 31 meter
 - 1 mm $\sim$ ddd° mm' **ss.sss ss''** (DMS) **.. (5-digits)**

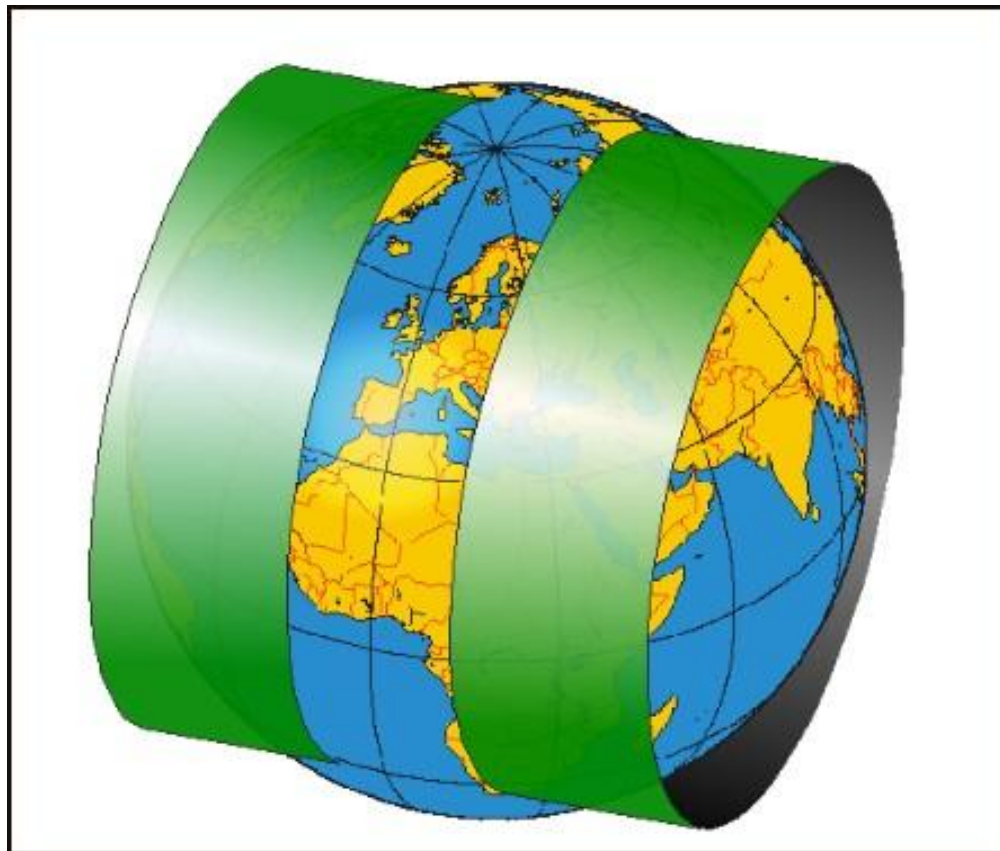
The Universal Transverse Mercator Projection (UTM)

การฉายแผนที่ (Map Projection)



One of these types is called Mercator

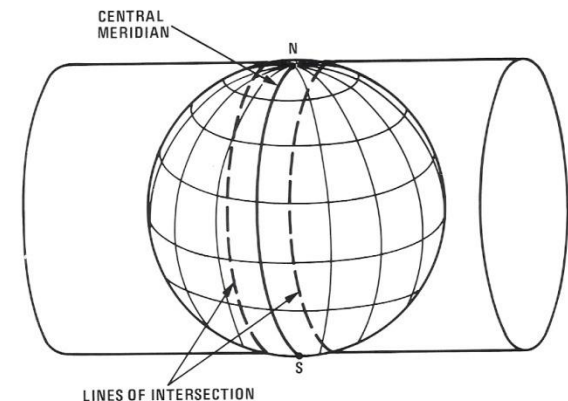
Universal Transverse Mercator (UTM)



UTM (Universal Transverse Mercator)

- Divided into sixty longitudinal zones
- Each zone is six-degrees in width extending three degrees on each side of the central meridian
- UTM is applicable between latitude 84N to 80S

1 degree = 110 km
1 minute = 1.8 km
1 second = 30 meter



UTM

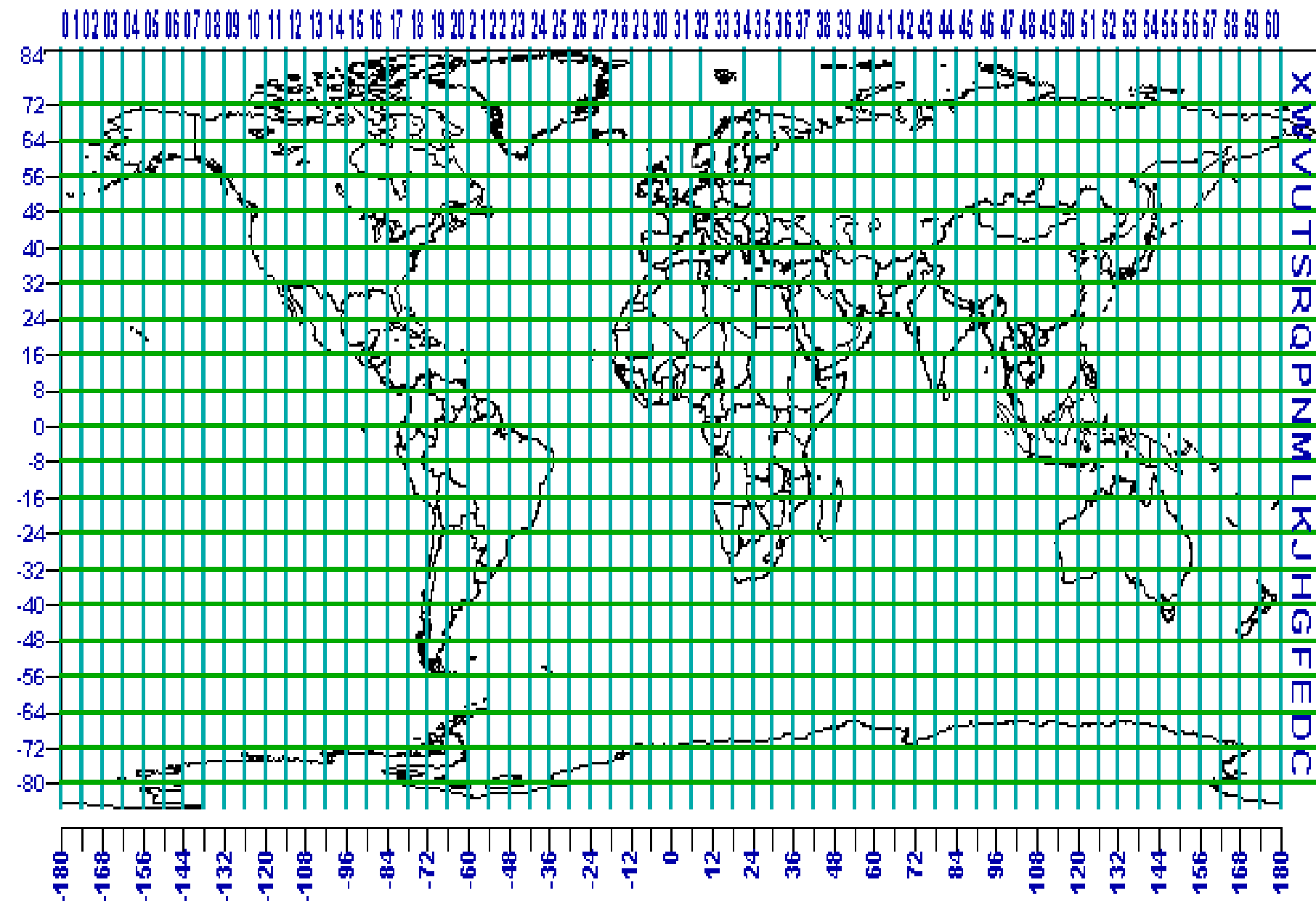
- Adopted by U.S. Army in 1947
- A conformal mapping, dividing Earth into 60 zones
- Each zone is 6 degree wide in longitude
- Origin is at the equator
- **False Northing**
 - for N-hemisphere 0 meter
 - For S-hemisphere 10,000,000 meter
- **False Easting = +500,000 meter for N/S hemisphere !!!**
- Scale Factor at central meridian = 0.9996 (1:2,500 or - 400ppm)
- Scale factor is half below 1.0010 or 1000 ppm! at the zone boundary along the equator

Scale Factor VS PPM

- PPM : Part-per-Million
 - 1 ppm = 1 mm in 1 kilometer
- Scale Factor
 - 0.9996 == -400 ppm
 - 1.0004 == +400 ppm
- UTM Zone:
 - Max PPM 1000 ??? pm

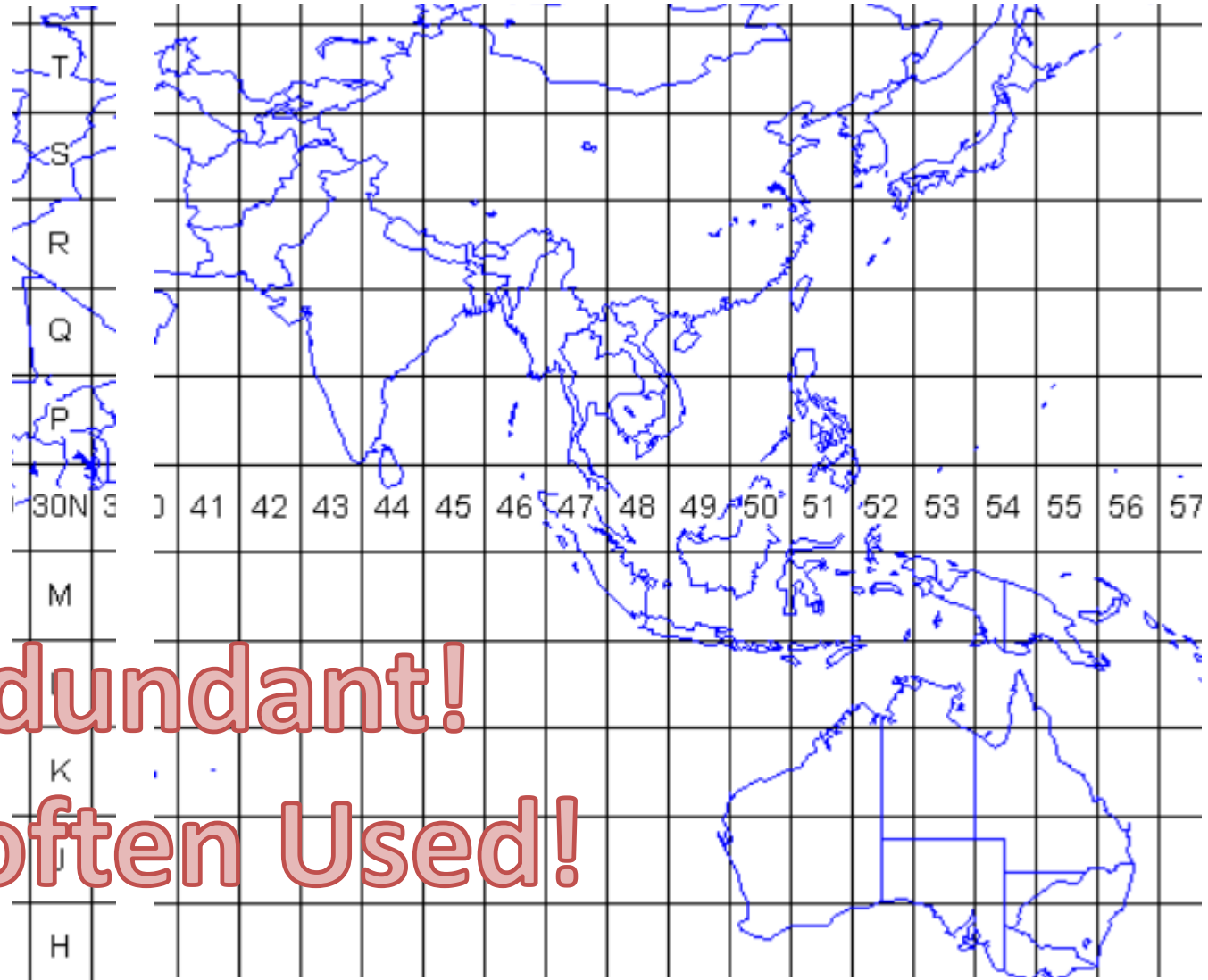
UTM Zone Numbers

UTM Zone Designators



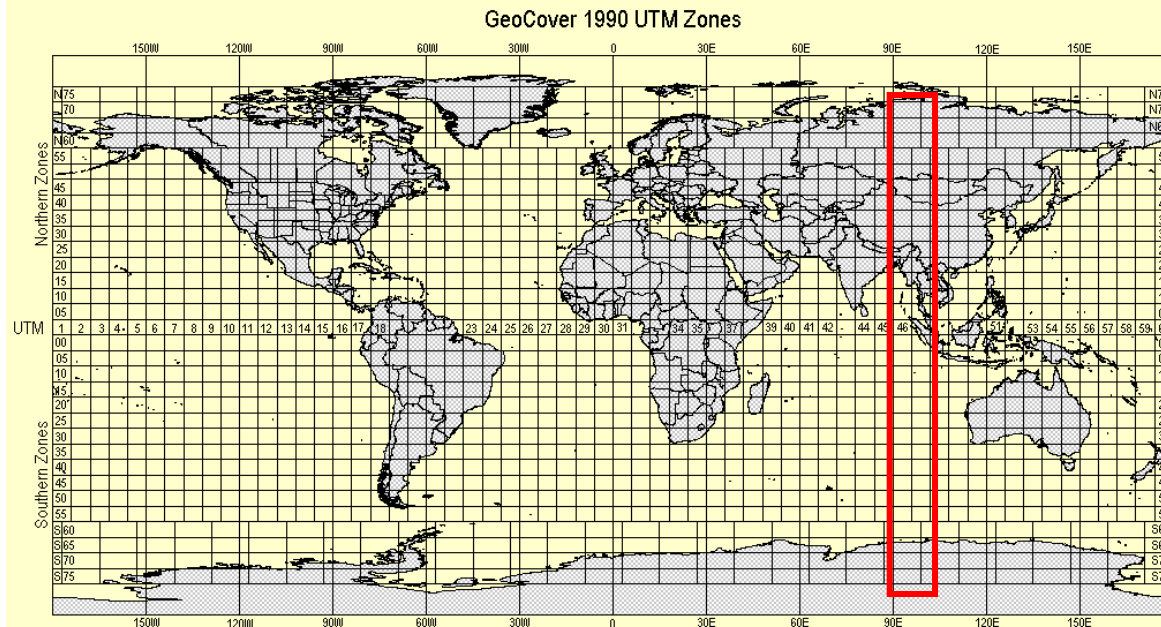
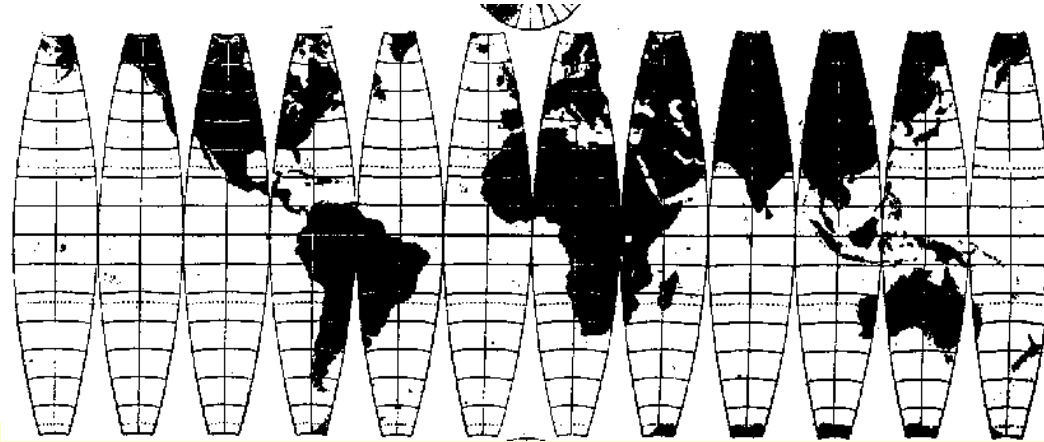
Universal Transverse Mercator (UTM) System

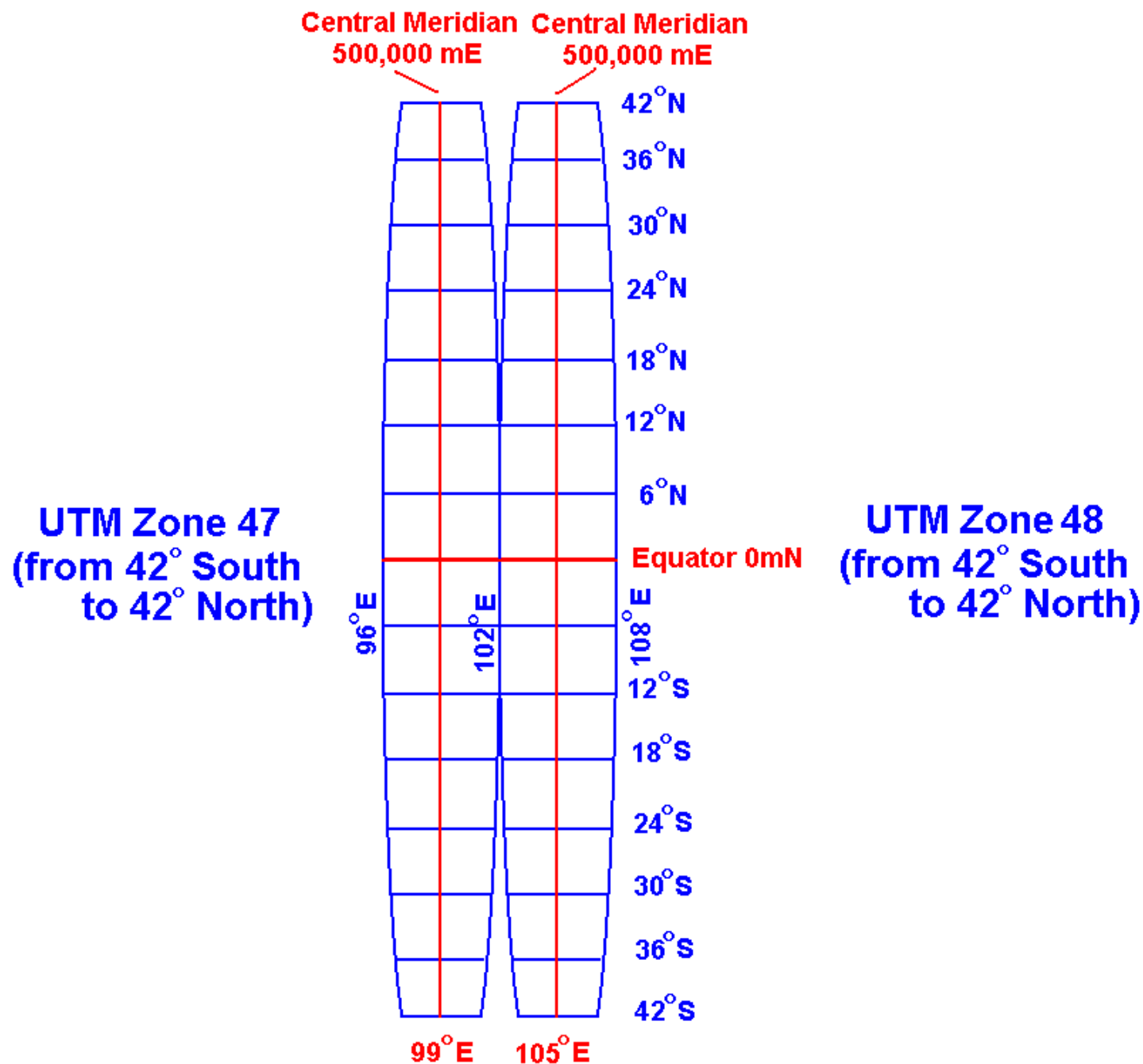
GZD : Grid Zone Designator



Redundant!
Not often Used!

Thailand is covered by UTM zone number 47 and 48

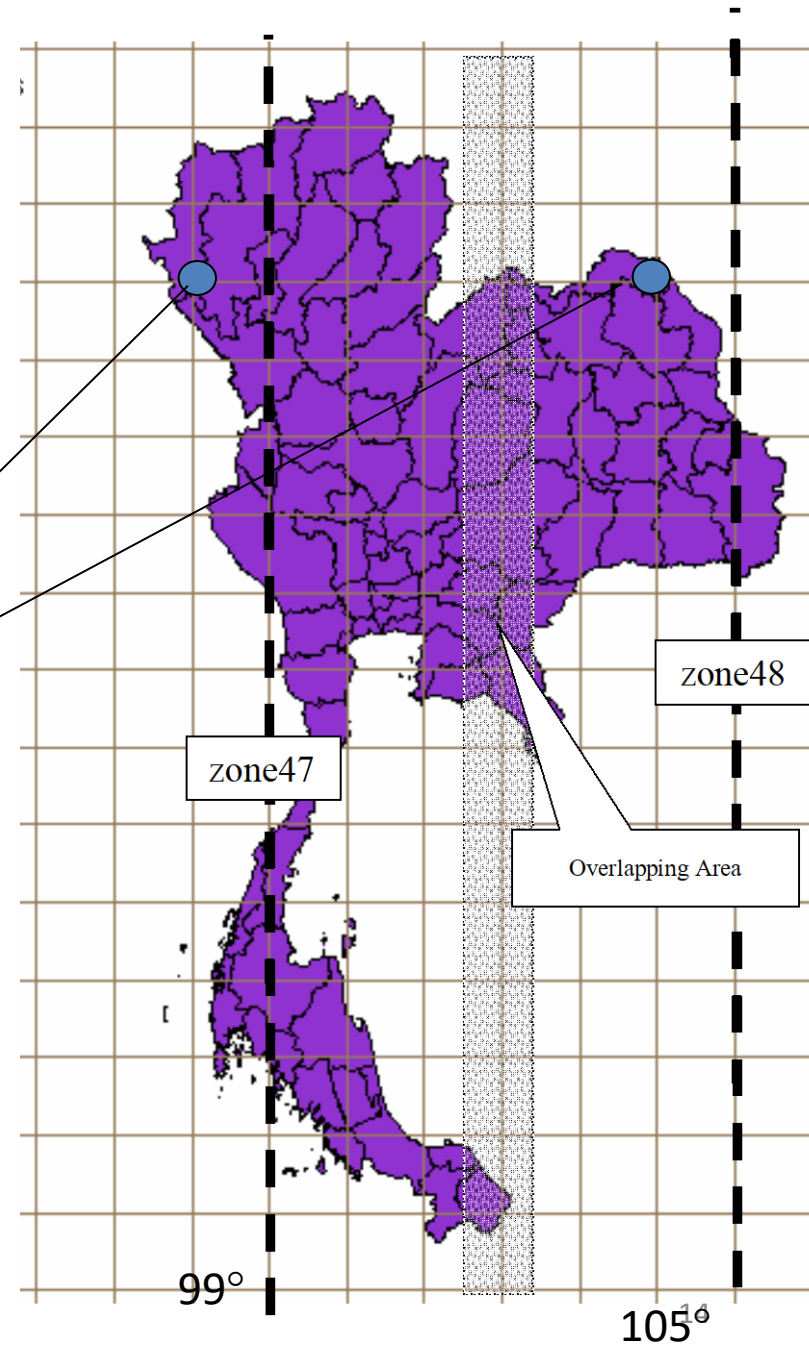




Common Pitfalls:

UTM zone 47 และ 48

- points at the same latitude and are the same distance from its central meridians will have the same coordinate
- There is often noting zone number in overlapping area between zone 47-48



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Map Projections— A Working Manual

By JOHN P. SNYDER

U.S. GEOLOGICAL SURVEY PROFESSIONAL PAPER 1395

Supersedes USGS Bulletin 1532



Library of Congress Cataloging in Publication Data

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(U.S. Geological Survey professional paper ; 1395)

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1. Map-projection--Handbooks, manuals, etc. I. Title. II. Series: Geological Survey professional paper ; 1395.
GA110.S577 1987 526.8 87-600250

For sale by the Superintendent of Documents, U.S. Government Printing Office
Washington, DC 20402

Transverse Mercator on Sphere

e.g. $R=6,371$ km

In an age of computers, it is much more satisfactory to use mathematical formulas. The rectangular coordinates for the Transverse Mercator applied to the sphere (Thomas, 1952, p. 6):

$$x = \frac{1}{2} R k_0 \ln [(1 + B)/(1 - B)] \quad (8-1)$$

or

$$x = R k_0 \operatorname{arctanh} B \quad (8-2)$$

$$y = R k_0 \{ \arctan [\tan \phi / \cos (\lambda - \lambda_0)] - \phi_0 \} \quad (8-3)$$

$$k = k_0 / (1 - B^2)^{1/2} \quad (8-4)$$

where

$$B = \cos \phi \sin (\lambda - \lambda_0) \quad (8-5)$$

(note: If $B = \pm 1$, x is infinite)

ระยะบนกริดแผนที่ = **ScaleFactor x** ระยะทางจริงบนทรงรี

Inverse Formulas

and k_0 is the scale factor along the central meridian λ_0 . The origin of the coordinates is at (ϕ_0, λ_0) . The Y axis lies along the central meridian λ_0 , y increasing northerly, and the X axis is perpendicular, through ϕ_0 at λ_0 , x increasing easterly.

The inverse formulas for (ϕ, λ) in terms of (x, y) :

$$\phi = \arcsin [\sin D / \cosh (x/Rk_0)] \quad (8-6)$$

$$\lambda = \lambda_0 + \arctan [\sinh (x/Rk_0) / \cos D] \quad (8-7)$$

where

$$D = y/(Rk_0) + \phi_0, \text{ using radians} \quad (8-8)$$

Rectangular coordinates for the sphere are shown in table 10. Only one octant (quadrant of a hemisphere) needs to be listed, since all other octants are identical except for sign change. See p. 268 for numerical examples.

Formulas for the Ellipsoid

For the ellipsoidal form, the most practical form of the equations is a set of series approximations which converge rapidly to the correct centimeter or less at full scale in a zone extending 3° to 4° of longitude from the central meridian. Beyond this, the forward series as given here is accurate to about a centimeter at 7° longitude, but the inverse series does not have sufficient terms for this accuracy. The forward series may be used with meter accuracy to 10° of longitude. (Many additional terms for use to 24° of longitude may be found in Army (1962).) Coordinate axes are the same as they are for the spherical formulas above. The for-

8. TRANSVERSE MERCATOR PROJECTION

mulas below are only slightly modified from those presented in standard references to provide mm accuracy at full scale (Army, 1973, p. 5–7; Thomas, 1952, p. 2–3). (See p. 269 for numerical examples.)

$$x = k_0 N [A + (1 - T + C) A^3/6 + (5 - 18T + T^2 + 72C - 58e'^2) A^5/120] \quad (8-9)$$

$$y = k_0 \{ M - M_0 + N \tan \phi [A^2/2 + (5 - T + 9C + 4C^2) A^4/24 + (61 - 58T + T^2 + 600C - 330e'^2) A^6/720] \} \quad (8-10)$$

$$k = k_0 [1 + (1 + C) A^2/2 + (5 - 4T + 42C + 13C^2 - 28e'^2) A^4/24 + (61 - 148T + 16T^2) A^6/720] \quad (8-11)$$

where k_0 = scale on central meridian (e.g., 0.9996 for the UTM projection)

$$e'^2 = e^2/(1 - e^2) \quad (8-12)$$

$$N = a/(1 - e^2 \sin^2 \phi)^{1/2} \quad (4-20)$$

$$T = \tan^2 \phi \quad (8-13)$$

$$C = e'^2 \cos^2 \phi \quad (8-14)$$

$$A = (\lambda - \lambda_0) \cos \phi, \text{ with } \lambda \text{ and } \lambda_0 \text{ in radians} \quad (8-15)$$

$$M = a[(1 - e^2/4 - 3e^4/64 - 5e^6/256 - \dots) \phi - (3e^2/8 + 3e^4/32 + 45e^6/1024 + \dots) \sin 2\phi + (15e^4/256 + 45e^6/1024 + \dots) \sin 4\phi - (35e^6/3072 + \dots) \sin 6\phi + \dots] \quad (3-21)$$

with ϕ in radians. M is the true distance along the central meridian from the Equator to ϕ . See equation (3-22) for a simplification for the Clarke 1866 ellipsoid.

$M_0 = M$ calculated for ϕ_0 , the latitude crossing the central meridian λ_0 at the origin of the x, y coordinates.

Note: If $\phi = \pm \pi/2$, all equations should be omitted except (3-21), from which M and M_0 are calculated. Then $x = 0$, $y = k_0(M - M_0)$, $k = k_0$.

Inverse Formulas

MAP PROJECTIONS—A WORKING MANUAL

$$e'^2 = e^2/(1-e^2) \quad (8-12)$$

$$C_1 = e'^2 \cos^2 \phi_1 \quad (8-21)$$

$$T_1 = \tan^2 \phi_1 \quad (8-22)$$

$$N_1 = a/(1-e^2 \sin^2 \phi_1)^{1/2} \quad (8-23)$$

$$R_1 = a(1-e^2)/(1-e^2 \sin^2 \phi_1)^{3/2} \quad (8-24)$$

$$D = x/(N_1 k_0) \quad (8-25)$$

To convert from tabular rectangular coordinates to ϕ and λ , it is necessary to subtract any “false easting” from x and “false northing” from y before inserting x and y into the inverse formulas. To convert coordinates measured on an existing map, the correct central meridian must be used for the Y axis on the Transverse Mercator, but the X axis may cross it perpendicularly at any latitude chosen by the user.

$$k = k_0[1 + (1 + e'^2 \cos^2 \phi)x^2/(2k_0^2 N^2)] \quad (8-16)$$

These formulas are somewhat more precise than those used to compute the State Plane Coordinate tables, which were adapted to use desk calculators of 30–40 years ago. Table 11 shows the variation of k with x .

To obtain UTM or SPCS coordinates, the appropriate “false easting” is added to x and “false northing” added to y after calculation using (8–9) and (8–10).

For the *inverse formulas* (Army, 1973, p. 6, 7, 46; Thomas, 1952, p. 2–3):

$$\phi = \phi_1 - (N_1 \tan \phi_1 / R_1)[D^2/2 - (5 + 3T_1 + 10C_1 - 4C_1^2 - 9e'^2)D^4/24 + (61 + 90T_1 + 298C_1 + 45T_1^2 - 252e'^2 - 3C_1^2)D^6/720] \quad (8-17)$$

$$\lambda = \lambda_0 + [D - (1 + 2T_1 + C_1)D^3/6 + (5 - 2C_1 + 28T_1 - 3C_1^2 + 8e'^2 + 24T_1^2)D^5/120]/\cos \phi_1 \quad (8-18)$$

where ϕ_1 is the “footpoint latitude” or the latitude at the central meridian which has the same y coordinate as that of the point (ϕ, λ) .

It may be found from equation (3–26):

$$\begin{aligned} \phi_1 = & \mu + (3e_1/2 - 27e_1^3/32 + \dots) \sin 2\mu + (21e_1^2/16 \\ & - 55e_1^4/32 + \dots) \sin 4\mu + (151e_1^3/96 + \dots) \sin 6\mu + (1097e_1^4/512 - \dots) \\ & \sin 8\mu + \dots \end{aligned} \quad (3-26)$$

where

$$e_1 = [1 - (1 - e^2)^{1/2}]/[1 + (1 - e^2)^{1/2}] \quad (3-24)$$

$$\mu = M/[a(1 - e^2/4 - 3e^4/64 - 5e^6/256 - \dots)] \quad (7-19)$$

$$M = M_0 + y/k_0 \quad (8-20)$$

with M_0 calculated from equation (3–21) or (3–22) for the given ϕ_0 .

For improved computational efficiency using series (3–21) and (3–26), see p. 19. From ϕ_1 , other terms below are calculated for use in equations (8–17) and (8–18). (If $\phi_1 = \pm\pi/2$, (8–12), (8–21) through (8–25), (8–17) and (8–18) are omitted, but $\phi = \pm 90^\circ$, taking the sign of y , while λ is indeterminate, and may be called λ_0 . Also, $k = k_0$.)

$$e'^2 = e^2/(1-e^2) \quad (8-12)$$

$$C_1 = e'^2 \cos^2 \phi_1 \quad (8-21)$$

$$T_1 = \tan^2 \phi_1 \quad (8-22)$$

$$N_1 = a/(1-e^2 \sin^2 \phi_1)^{1/2} \quad (8-23)$$

$$R_1 = a(1-e^2)/(1-e^2 \sin^2 \phi_1)^{3/2} \quad (8-24)$$

$$D = x/(N_1 k_0) \quad (8-25)$$



Double precision is needed !

UTM False Origin

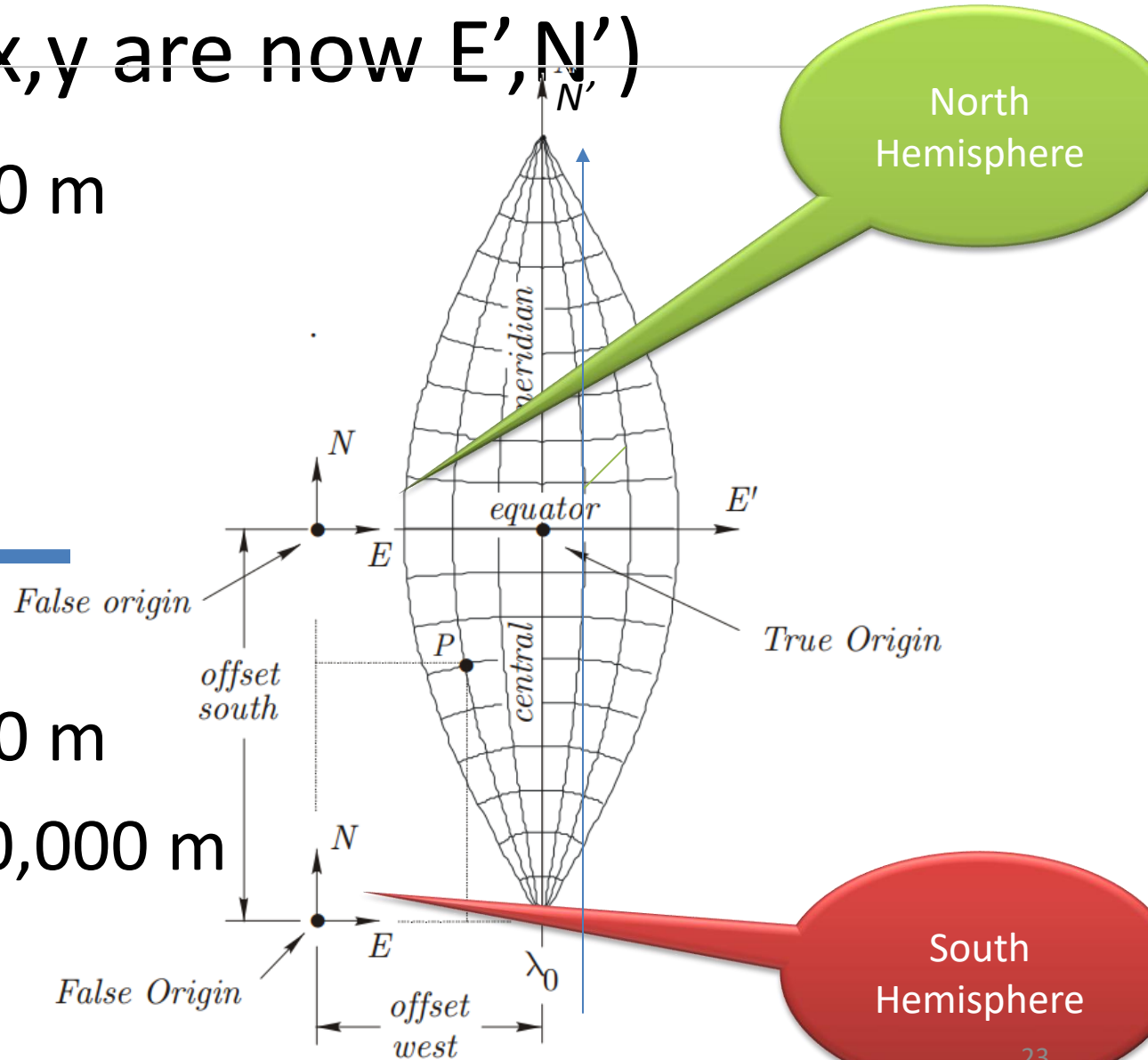
(x,y are now E',N')



The diagram illustrates the UTM False Origin. It shows a coordinate system with a horizontal axis labeled 'E' (Easting) and a vertical axis labeled 'N' (Northing). A point is marked with a crosshair and labeled 'N'' (Northing). A horizontal line is drawn through the point, and a vertical line is drawn through the point, intersecting at the point. The text 'UTM False Origin' is written above the diagram, and '(x,y are now E',N\'' is written below the diagram.

- $E = E' + 500,000 \text{ m}$
- $N = N'$

- $E = E' + 500,000 \text{ m}$
- $N = N' + 10,000,000 \text{ m}$



A FRESH LOOK AT THE UTM PROJECTION: Karney-Krueger equations

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³ Princeton, N.J., USA.

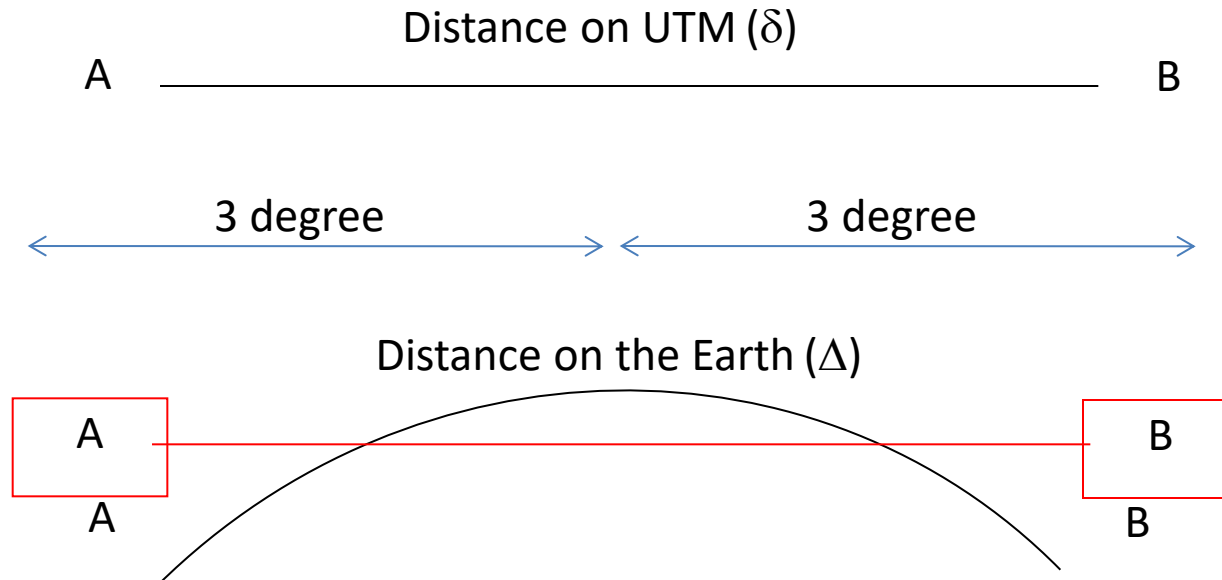
email: rod.deakin@rmit.edu.au

Presented at the Surveying and Spatial Sciences Institute (SSSI) Land Surveying Commission National Conference,
Melbourne, 18-21 April, 2012

UTM SCALE FACTOR

Point Scale Factor (PSF)

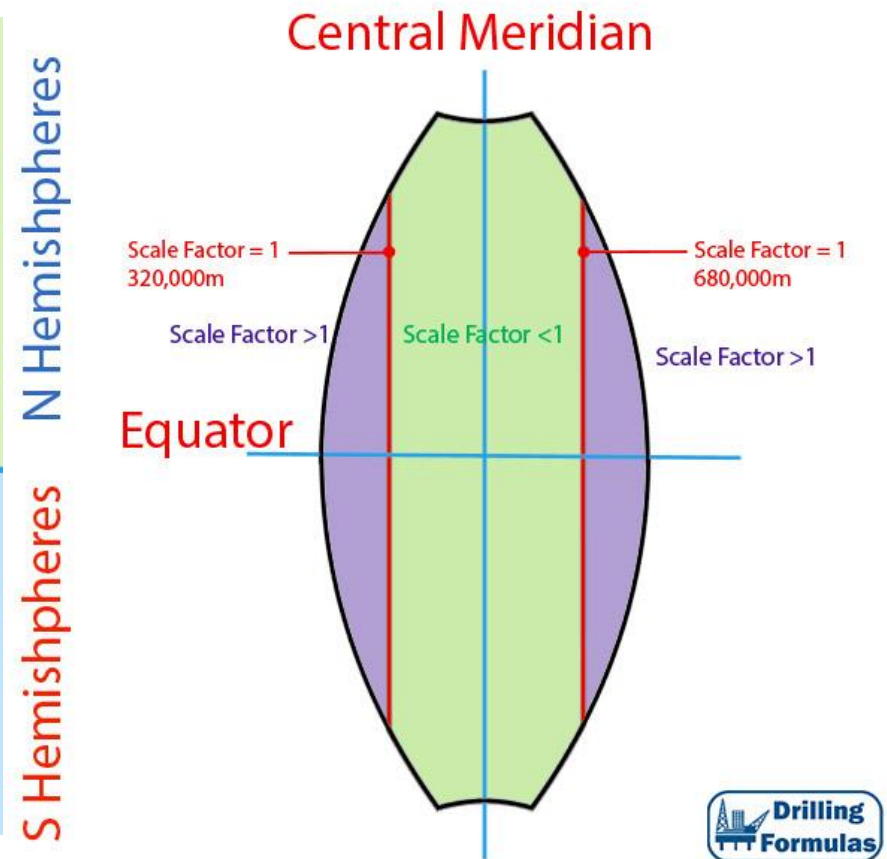
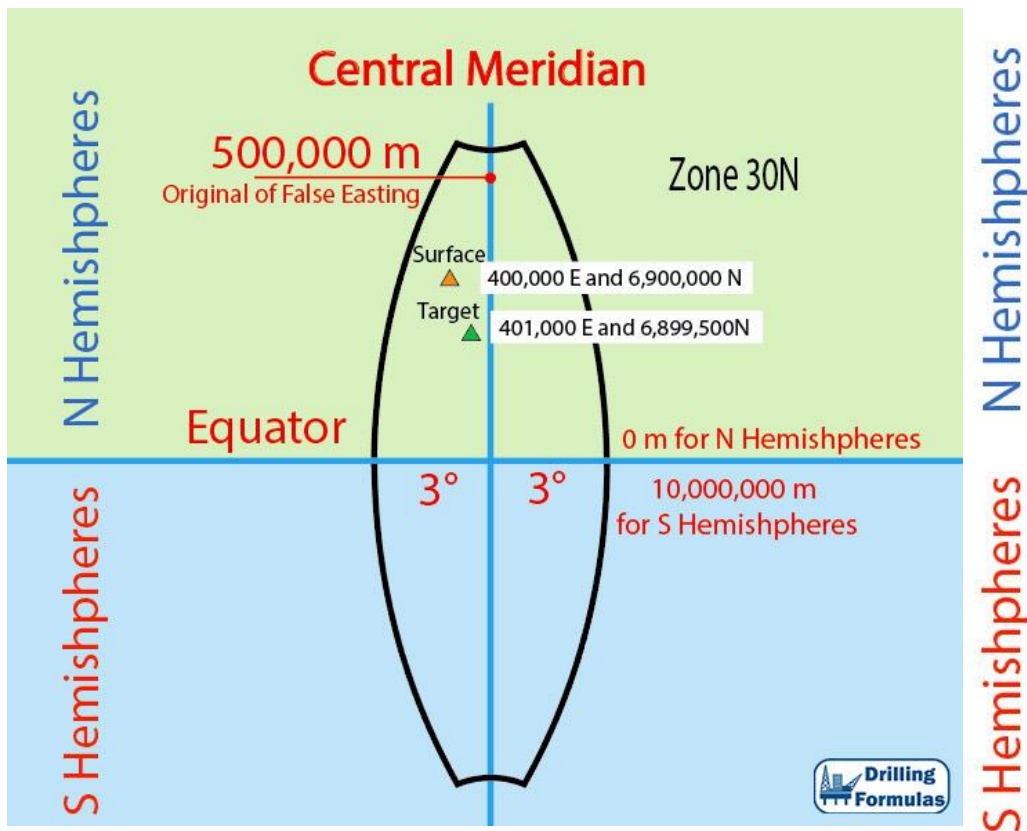
- scale factor = δ / Δ
UTM scale factor
= 0.9996 ... 1.0004



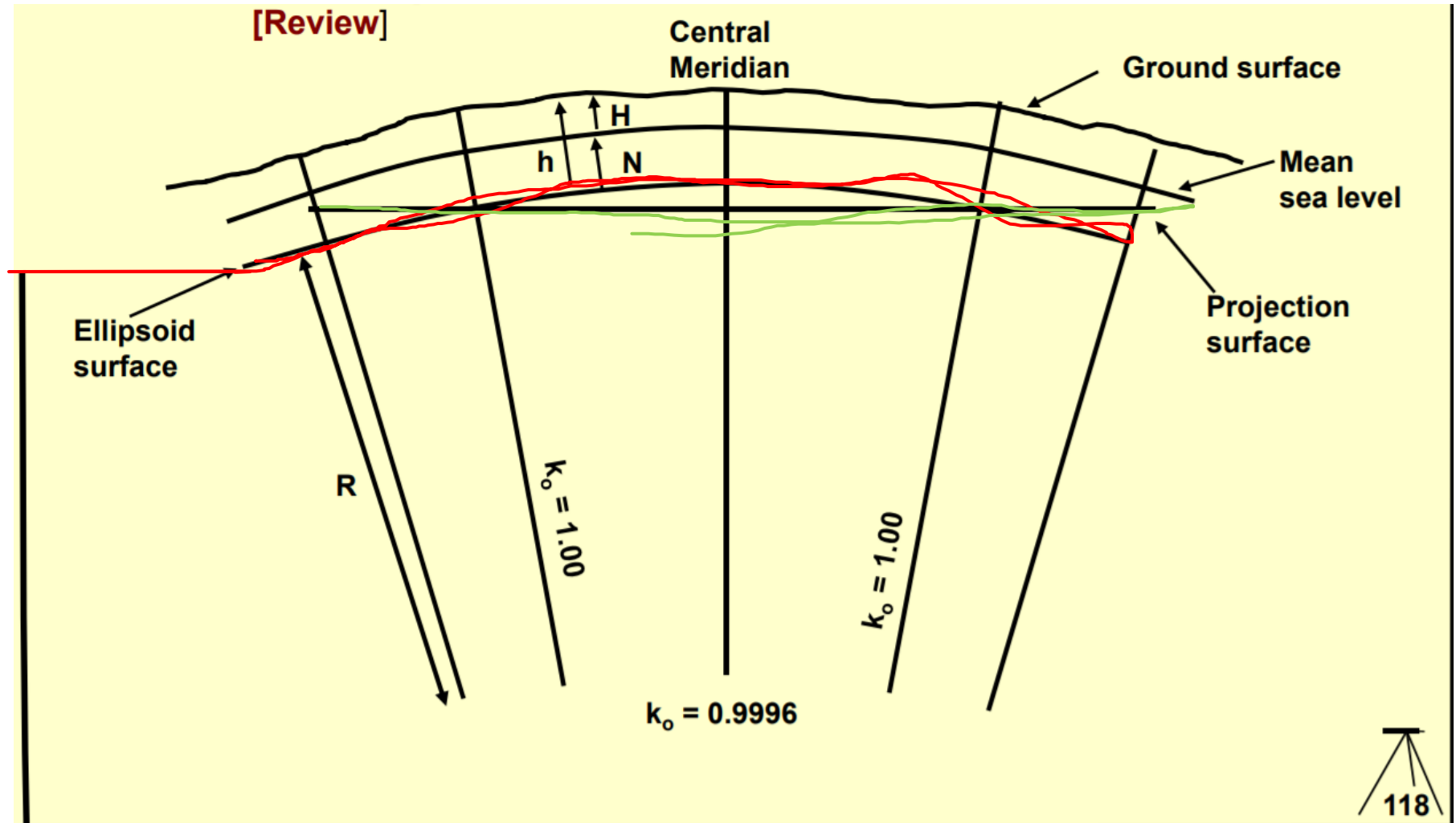
Scale factor on UTM

- Ranging from 0.9996 to 1.0004
- Result in scale or distance discrepancy less than 1 to 2,500
- E.g. a distance of 2,500 meter obtained on UTM map will differ from the actual distance on the Earth less than 1 meter.
- E.g. calculated distance of 1 kilometer on UTM map, it result errors on Earth 0.4 meter
- However this relation is calculable via mathematic formulae.

UTM Explanation

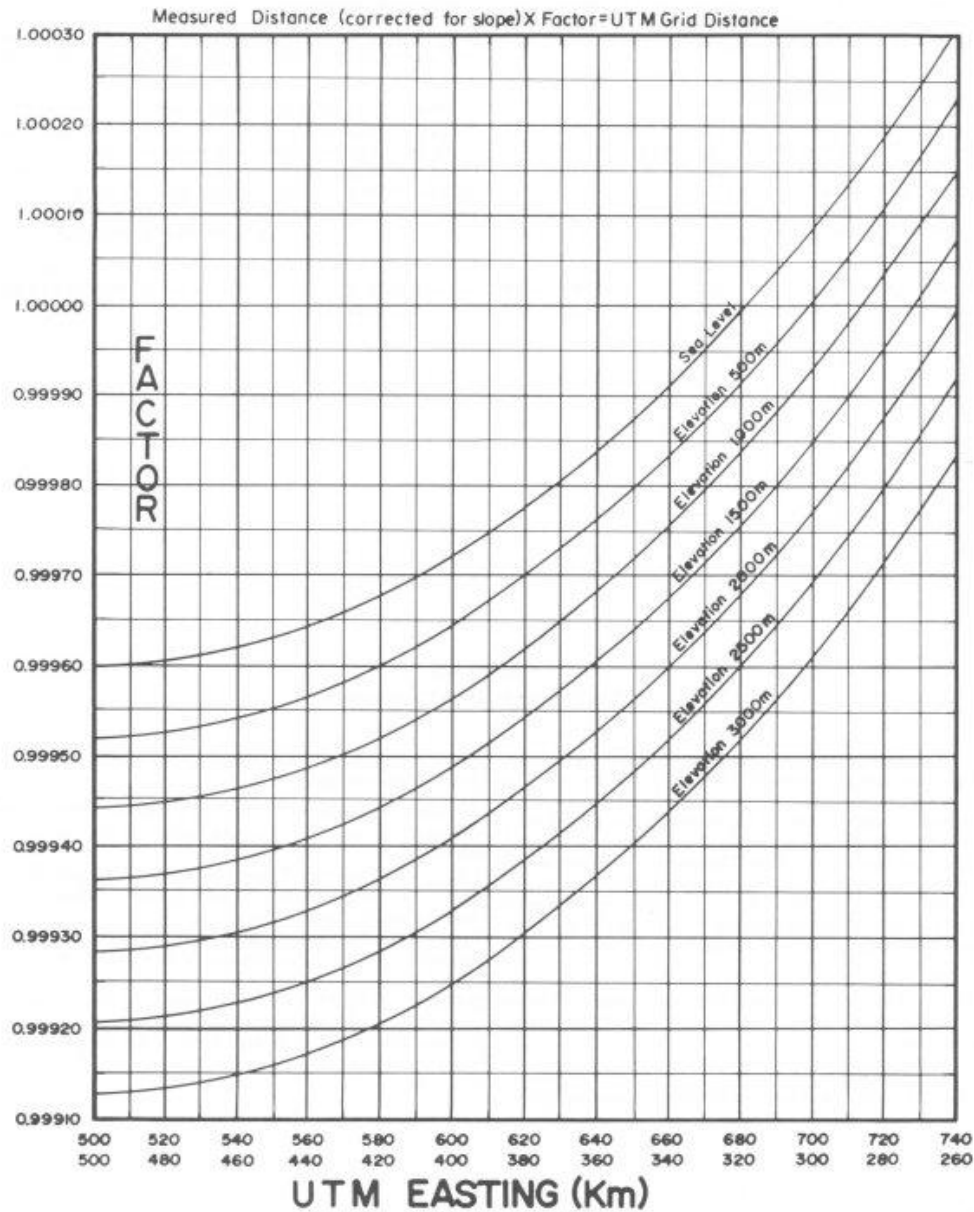


[Review]



FACTOR FOR REDUCING HORIZONTAL DISTANCES TO UTM GRID DISTANCES

(AT DIFFERENT ELEVATIONS AND UTM EASTINGS)



GeographicLib/GeoConvert

- Geodetic to UTM
- `>geoconvert -u -p 3 --input-string "14.00000N
100.00000E"`
- **47n 607995.652 1547954.320**
- Geodetic to UTM meridian convergence and scale factor
- `>geoconvert -c -p 6 --input-string "14.00000N
100.00000E"`
- **0.24194546725 0.9997442633311**

http://www.ga.gov.au/geodesy/datum s/redfearn_geo_to_grid.jsp

Updated: 21 May 2015

Geodetic Calculations - Redfearn's Formulae, Geographic to Grid

Given Latitude and Longitude, calculate Easting, Northing, Zone for a Universal Transverse Mercator (UTM) projection, Grid convergence and Point Scale Factor.

- ▶ The calculations are performed using the GRS80 ellipsoid which is used for Australia's new coordinate system ([the Geocentric Datum of Australia - GDA](#)) and is also compatible with the global coordinate system (WGS84) ($a = 6,378,137.0$ metres; $1/f = 298.25722210$)
- ▶ Redfearn's formulae are used (Redfearn, Empire Survey Review, No. 69, 1948) to convert between latitude & longitude and easting, northing & zone for a UTM projection, such as the Map Grid of Australia (MGA) used in conjunction with the GDA. These formulae are accurate to better than 1mm in any zone.
- ▶ Latitude must be between 0° and $+84^\circ/-80^\circ$. South Latitude is negative (eg $-35^\circ 55' 56.12''$).
- ▶ Longitude must be between 0° and $\pm 180^\circ$. West Longitude is negative (eg $-148^\circ 56' 25.12''$).
- ▶ **NOTE:** Given Easting, Northing and Zone for a UTM projection, calculate Latitude, Longitude, Grid convergence and Point Factor use the [Grid to Geographic version of Redfearn's Formula](#).

Point Name: eg Flinders Peak (Optional - for display only)

Latitude: [°] ['] ["] eg $-37^\circ 39' 10.15610''$

Longitude: [°] ['] ["] eg $143^\circ 55' 35.38390''$

GDA Redfearn Calculation Results (Geographic to Grid)

Site Name:	BKK	GDA-MGA:	(UTM with GRS80 ellipsoid)
Zone:	47		
Easting:	607995.652	Northing:	11547954.320
Latitude:	14 ° 0 ' 0.00000 "	Longitude:	100 ° 0 ' 0.00000 "
Grid Convergence:	0 ° -14 ' 31.00 "	Point Scale:	0.99974426

[Another calculation](#)

[Comments & Suggestions](#)

Classwork 6-2 : Scale Factor for Thailand

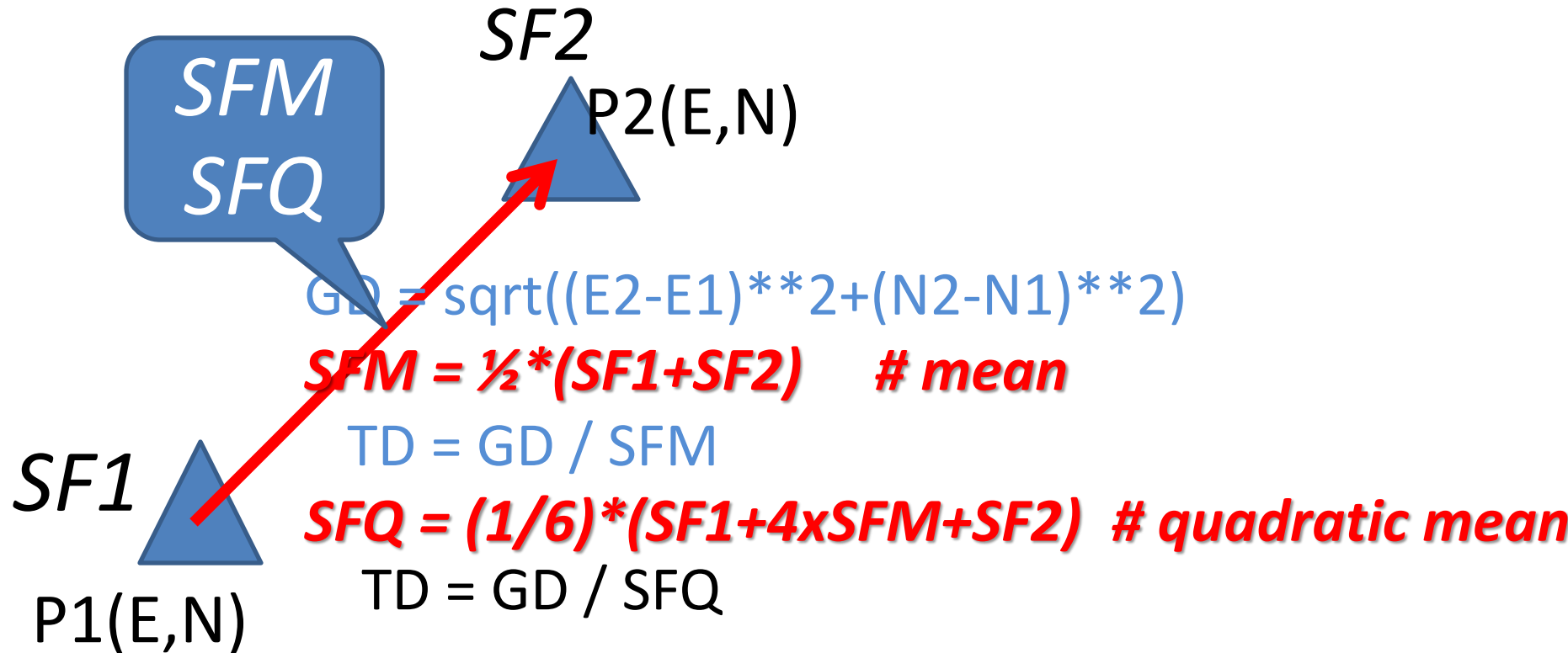
- ให้นิสิตอ่านค่าพิกัดที่ “มุมสุด **extreme**” ของโซน 47 และโซน 48 ที่ครอบคลุมประเทศไทย
- ให้พล็อตแผนที่แสดงจุด **extreme** ด้วยระบบพิกัดภูมิศาสตร์
- สำหรับ โซน 47 7 จุด (ที่ **cm=99** จะมี 2 จุด)
- สำหรับ โซน 48 7 จุด (ที่ **cm=105** จะมี 2 จุด)
- แล้วใช้ **geographic/geoconvert** คำนวณค่าสเกลแฟกเตอร์ และคำนวณเป็น **ppm** (part-per-million / mm per kilometer)
- สรุปว่าประเทศไทยมีความสเกลแฟกเตอร์ หรือ **ppm** สูงสุดที่ใด

GRID AND TRUE DISTANCE

UTM : True Dist & Grid Dist

Line Scale Factor (LSF)

- Point-Scale Factor (PSF) → LSF

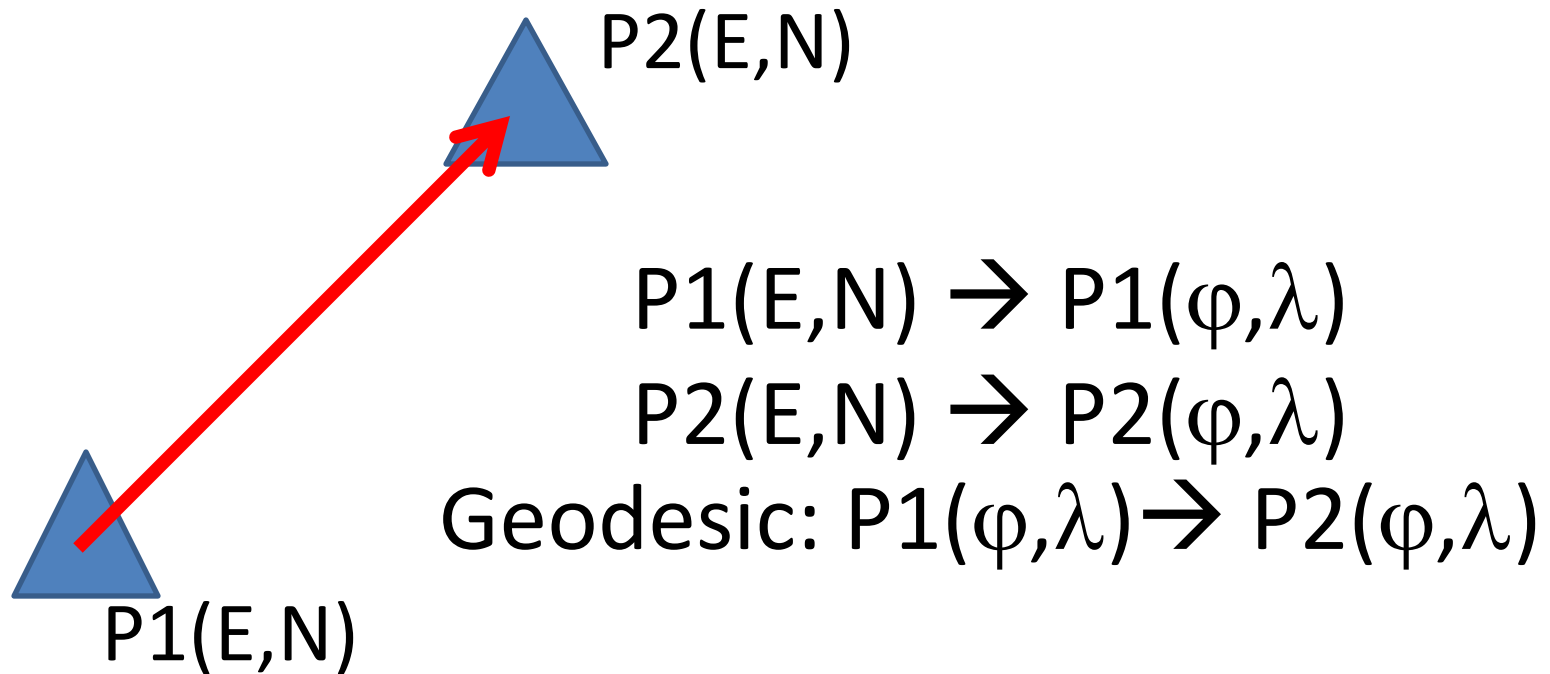


True Dist or Geodesic Dist (alternative, computing intensive ?)

- True Distance from $(E1, N1) \rightarrow (E2, N2)$?
- Convert $(E1, N1)$ to $(lat1, lng1)$
- Convert $(E2, N2)$ to $(lat2, lng2)$
- Compute 'geodesic length' from $(lat1, lng1)$ to $(lat2, lng2)$

Geodesic & Grid Dist

- Direct method / geodetic traverse



ClassWork 6-4 UTM Scale Factor

- นำค่าพิกัด @@-L ไปยังปลายทาง @@-R คำนวณระยะทางจริง (Geodesic Dist) ด้วย inverse problem คำสั่ง `geographiclib/geosolve` นำผลระยะทางที่ได้มาเปรียบเทียบกับ `grid dist` กรณีใช้ สเกลแฟกเตอร์ที่ ต้น ปลาย และกลางทาง
- ความต่างระยะทางเป็นเท่าใดบ้าง อภิปราย
- การคำนวณด้วย คำสั่ง `geographiclib` ให้พิมพ์คำสั่ง แบบ เบ็ดเสร็จบรรทัดเดียว มีออฟชั่น `-input-string` “ ... ”
- ความละเอียดในการคำนวณให้ใช้ทศนิยมละเอียดถึง “มิลลิเมตร”

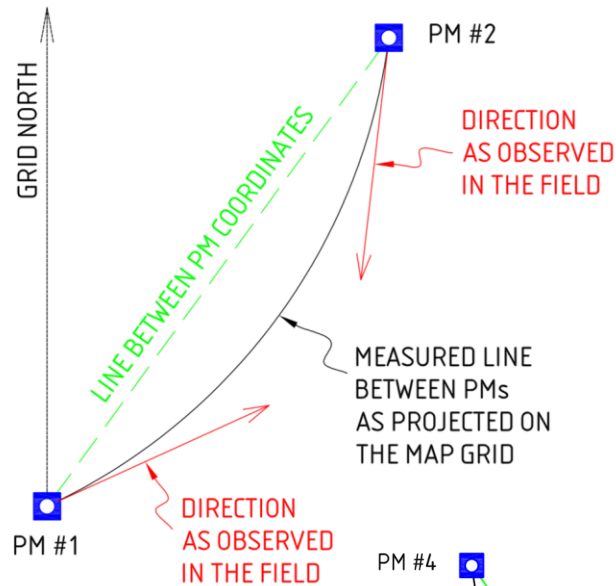
ClassWork 6-5 : Line Scale Factor (LSF)

- ให้นักนิสิตเลือกสนามบินตามหมายเลข CR58 ตรวจสอบอักษรย่อ 3 ตัว ค้นหาด้วย ‘@@@ airport coordinate’ คนละหนึ่งแห่ง เปิด google maps แล้วเลือก “ทางวิ่งหลัก” ที่ยาวที่สุด 1 แห่ง
- ทำการสเกตส์ ขนาดและทิศทางวิ่ง แล้วอ่านพิกัดต้นทาง @@-L ไปยังปลายทาง @@-R ด้วย ‘What’s here ?’ เช่น 43.674403, -79.663716 และ 43.694214, -79.634931
- ให้นักนิสิตคำนวณพิกัดทั้งสอง ไปเป็น Zone/UTM geographiclib/geoconvert แล้วคำนวณระยะทางระหว่างจุดทั้งสองด้วย Pythagorus เรียกระยะนี้ว่า Grid Dist (GD)
- ให้คำนวณค่าสเกลแฟกเตอร์ ของจุดทั้งสอง SF1,SF2 และค่าเฉลี่ย SFM,SFQ สเกลแฟกเตอร์ควรมีเลขนัยสำคัญกี่ตัว จริงจะพอเพียงสำหรับการคำนวณละเอียดถึง “มิลลิเมตร”
- จัดทำตารางแสดงการแปลงระยะทางกริด (GD) ไปเป็นระยะทางจริง (True Dist: TD) ด้วย SF1, SF2,SFM, SFQ และผลต่างเทียบกับระยะทางจริงเทียบกับเส้นยี่สิบเอ็ดทิศ คำนวณด้วย geographiclib/geodsove -i ผลต่างที่ได้ให้แสดงเป็น หน่วย มิลลิเมตร

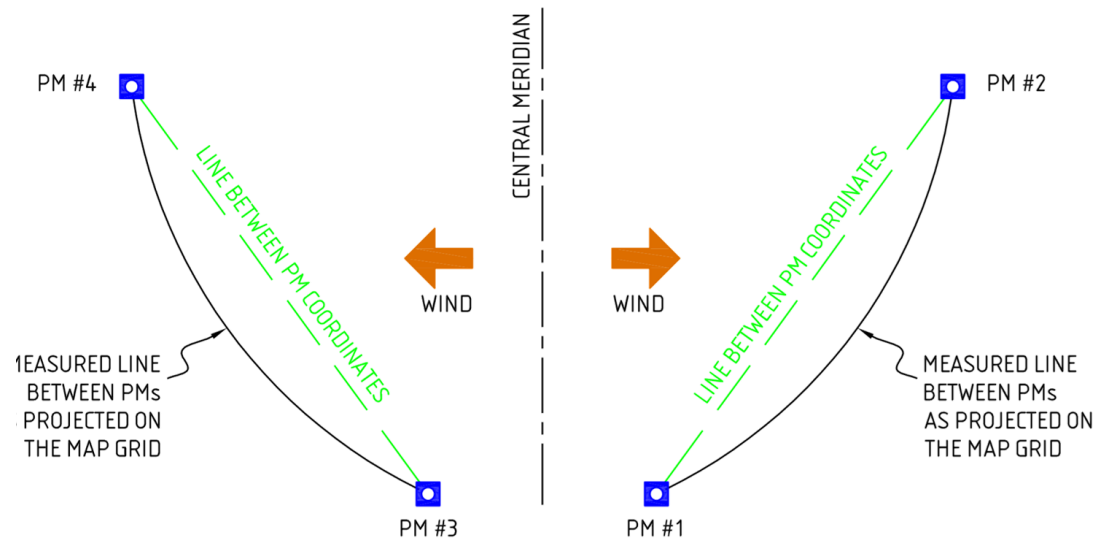
Arc-to-Chord Correction

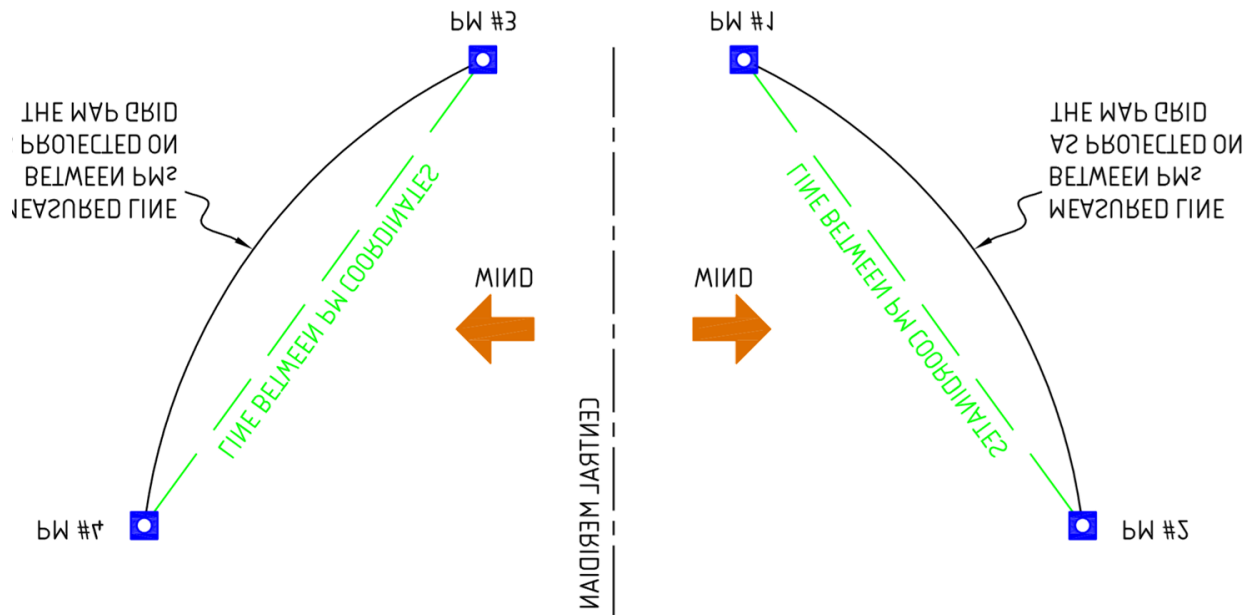
MAP GRID PROJECTION PLANE

CENTRAL MERIDIAN

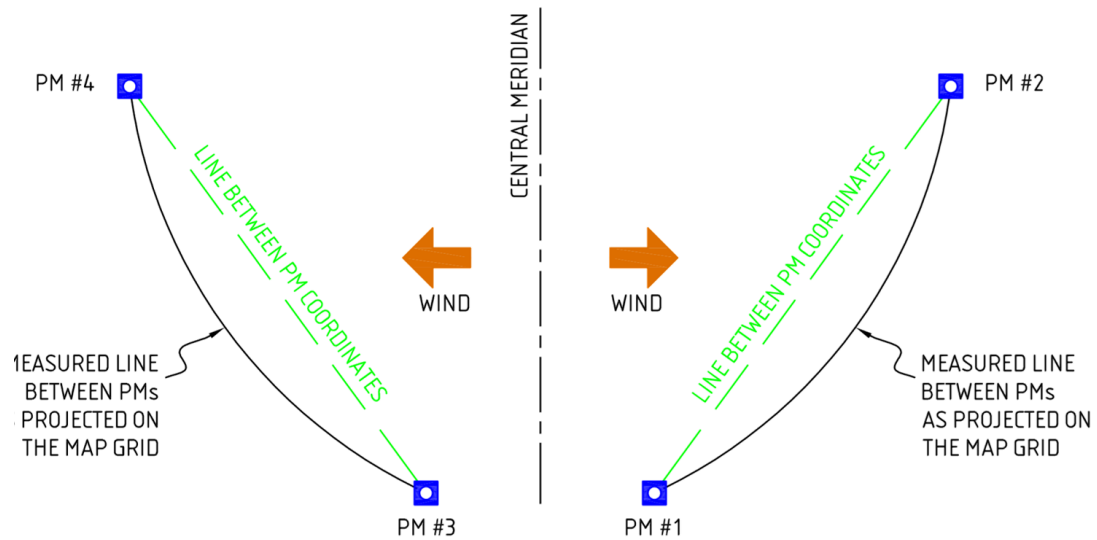


MAP GRID PROJECTION PLANE

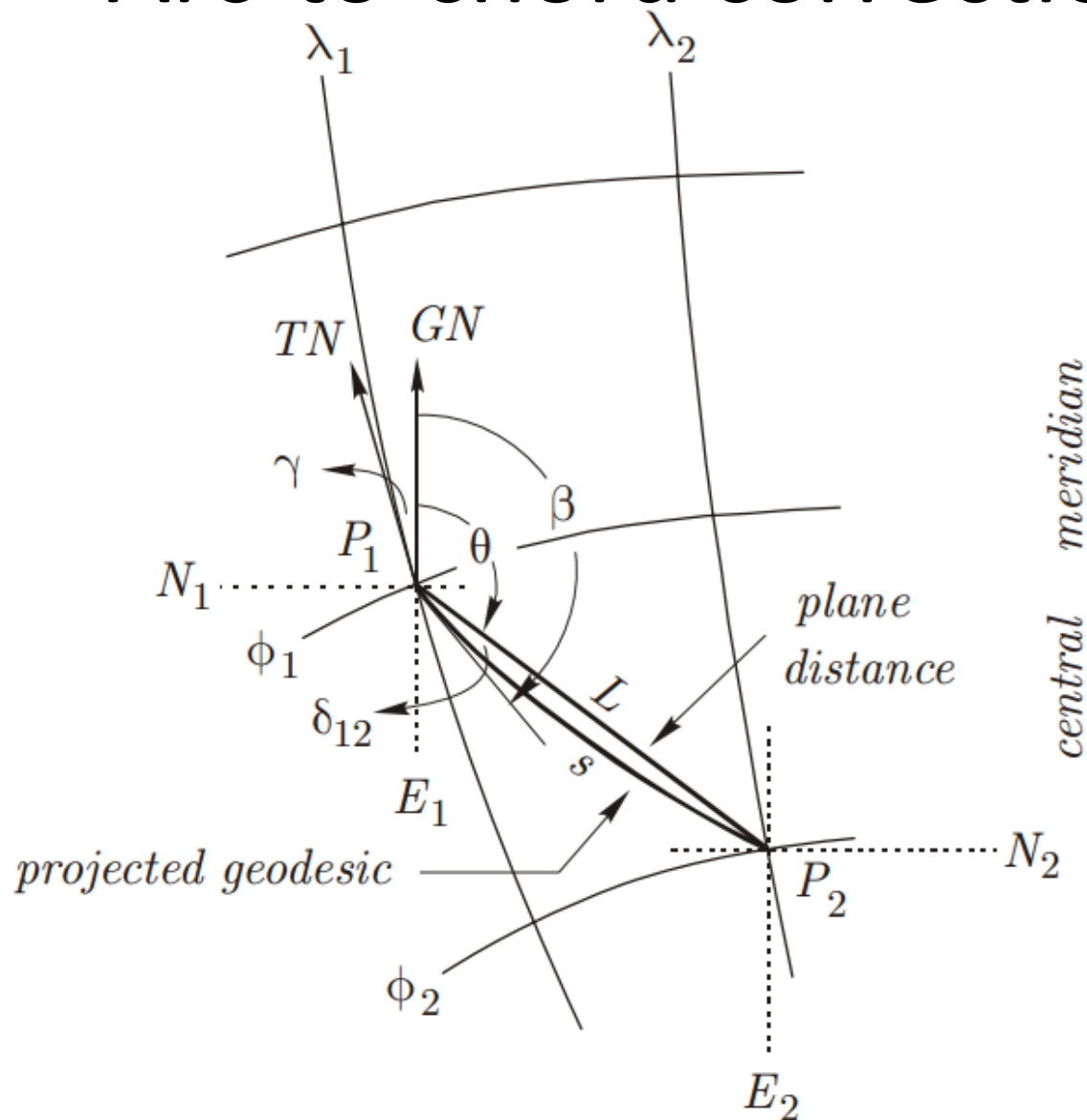




MAP GRID PROJECTION PLANE



Arc-to-chord correction



Arc-to-chord correction

The small angle between the tangent to the projected geodesic at P_1 and the straight line joining P_1 and P_2 is the *arc-to-chord* correction δ_{12} and is given by

$$\delta_{12} = -\frac{(N_2 - N_1)(E'_2 + 2E'_1)}{6r_m^2} \left\{ 1 - \frac{(E'_2 + 2E'_1)^2}{27r_m^2} \right\} \quad (7)$$

omitted as its effect is negligible. For a line 100 km in length running north and south on a zone boundary the error in neglecting this term is about 0.08" (NMC 1985).

GeographicLib\GeodSolve

- >GeodSolve.exe -i -p 6 --input-string "14 100
15 101"

True Azimuth
At P1

- 44.12901925751

True Azimuth
At P2

44.37941483092

True Dist

154472.517901

Online geodesic calculations using the [GeodSolve](#) utility

Geodesic calculation:

- Inverse: $lat1\ lon1\ lat2\ lon2 \rightarrow azi1\ azi2\ s12$
- Direct: $lat1\ lon1\ azi1\ s12 \rightarrow lat2\ lon2\ azi2$

Input (ex. «40,6 -73,8 49°01'N 2°33'E» [inverse], «40d38'23"N 073d46'44"W 53d30'

14. 100. 15. 101.

Output format: ● decimal degrees ● degrees minutes seconds
Heading at point 2: ● forward azimuth ● back azimuth
Longitude: ● reduce to $[-180^\circ, 180^\circ]$ ● unroll
Output precision:
Equatorial radius: meters
Flattening:

Select action:

Geodesic (input in black, output in blue):

ellipsoid (a f) = 6378137 1/298.257223563 (WGS84)
status = OK

lat1 lon1 fazi1 (°) = 14.00000000 100.00000000 44.12901926
lat2 lon2 fazi2 (°) = 15.00000000 101.00000000 44.37941483
s12 (m) = 154472.518

a12 (°) = 1.39202611
m12 (m) = 154457.328
M12 M21 = 0.9997049987 0.9997050098
S12 (m^2) = 176642939885

การคำนวณปรับแก้วงรอบ

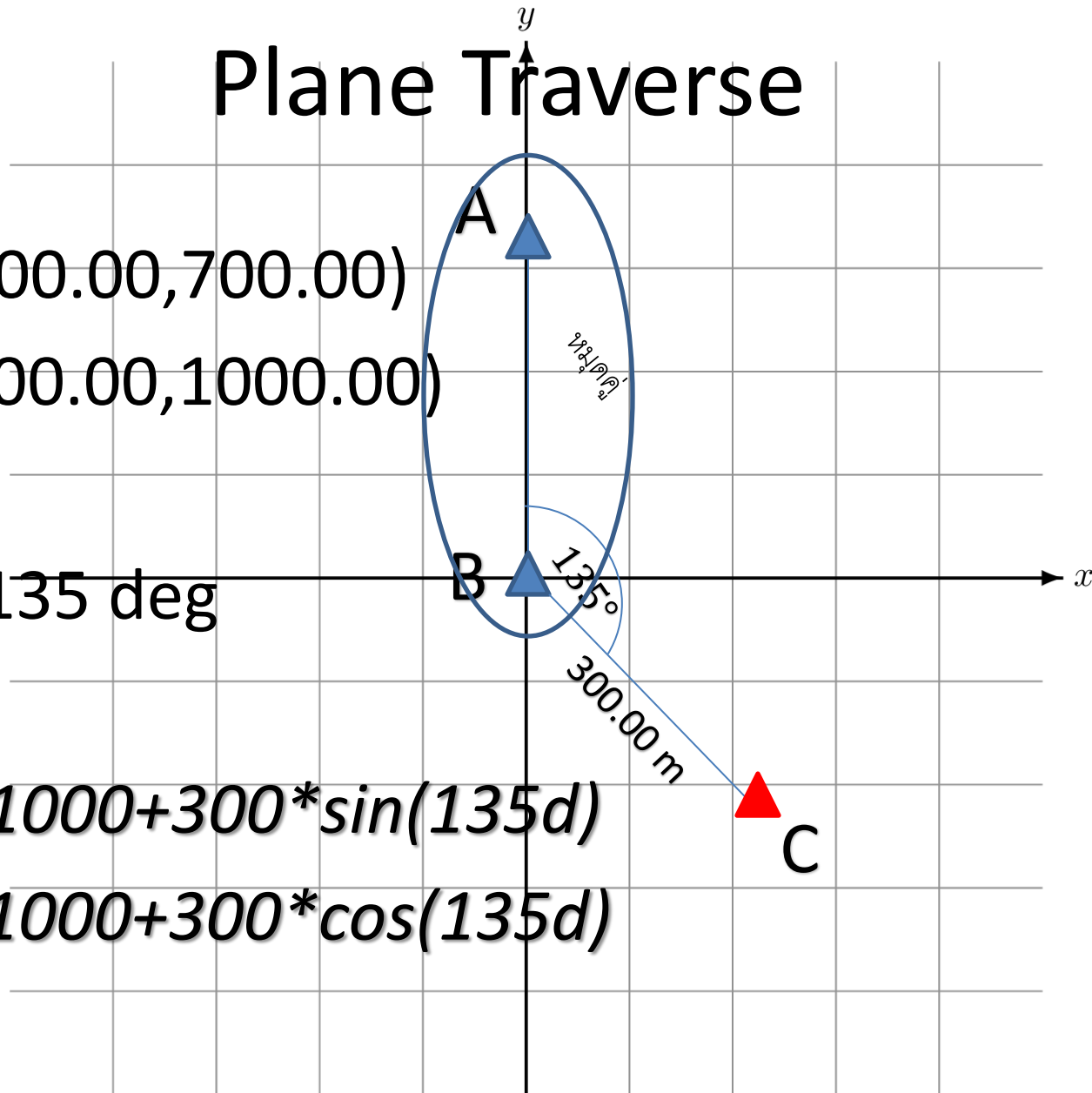
Traverse Adjustment Computation

Plane Traverse

- A (1000.00,700.00)
- B (1000.00,1000.00)

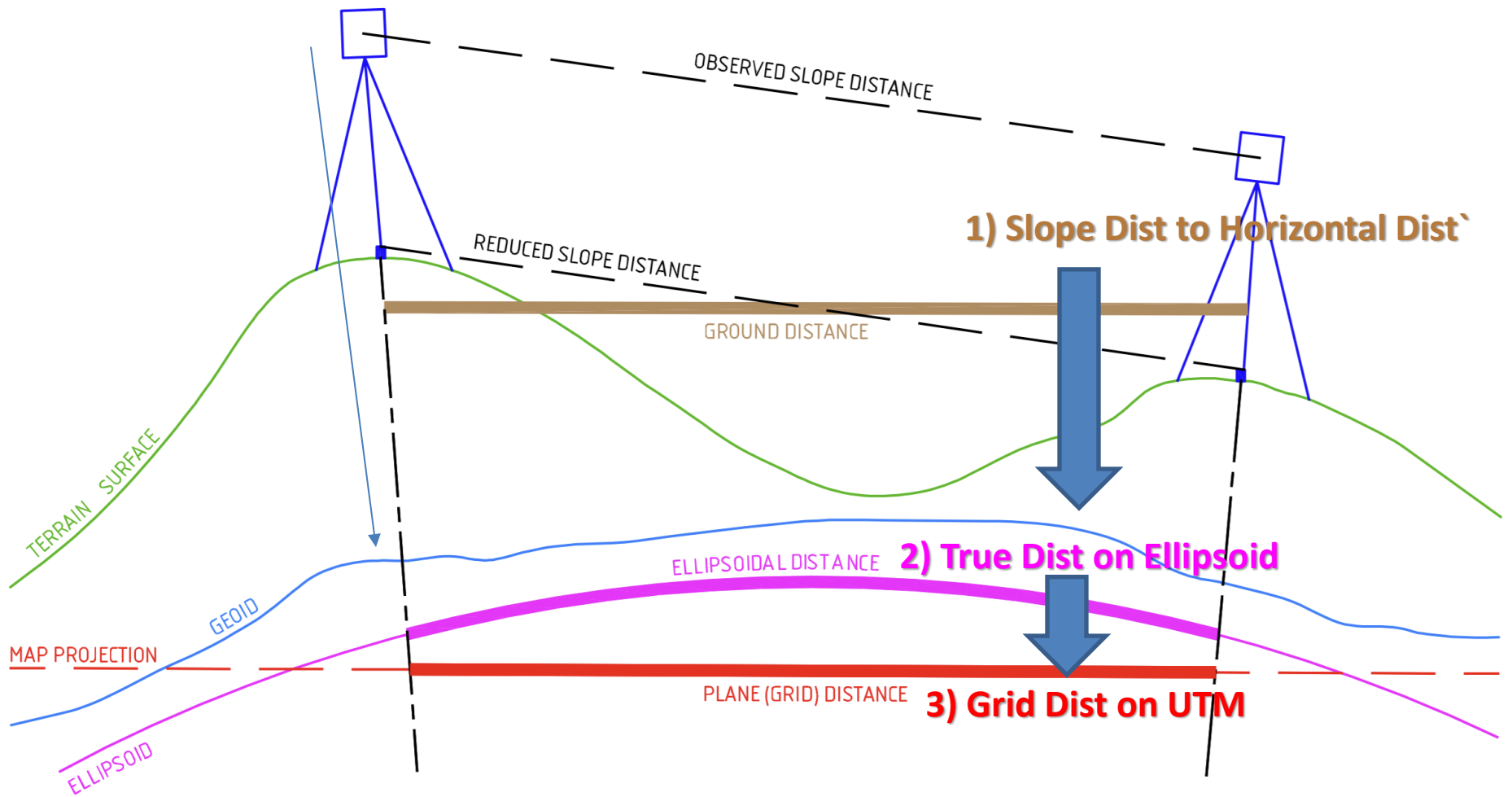
- Az = 135 deg

- $CX = 1000 + 300 * \sin(135^\circ)$
- $CY = 1000 + 300 * \cos(135^\circ)$



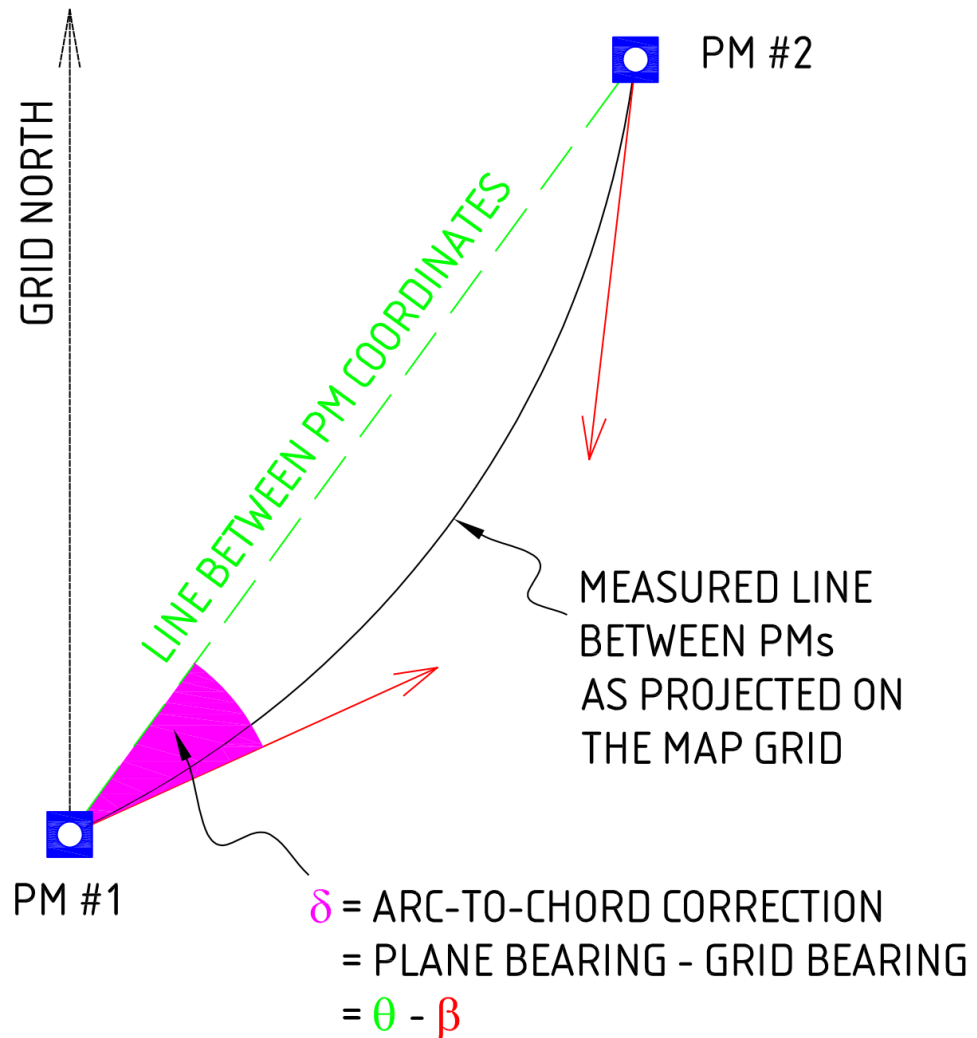
การปรับแก้วงรอบ

- ปรับแก้วงรอบบนระนาบยูทียีเอ็ม (plane traverse)
 - มุมภายในรูปหลายเหลี่ยม $(n-2)*180$
 - มุมรังวัดได้ คือผลต่างทิศทาง และมีค่าแก้ arc-to-chord อยู่ห่างจาก CM. อาจมีค่ามากถึง 30"
 - ยังคงต้องปรับแก้ คือ ความยาวระยะรังวัด ลงสูง ระยะทางบนแผนที่ (CSF=PSF*HSF)
- ปรับแก้วงรอบบนพื้นผิวทรงรี (geodetic traverse)
 - มุมรังวัดได้ คือผลต่างทิศทาง geodetic azimuth คำนวณได้จาก direct & indirect geodetic problem
 - ใช้ระยะทางจริงบนพื้นผิวทรงรี (geodesic length)
 - การคำนวณง่าย และลดความผิดพลาดเครื่องหมาย บวก/ลบ
 - ต้องการซอฟต์แวร์ที่สามารถ retracing ได้ เช่น Geographiclib/GeoSolve
 - คำนวณ geodetic azimuth สะสม จนไปบรรจบ geodetic azimuth สุดท้าย



Arc-to-chord correction

CENTRAL MERIDIAN



CENTRAL MERIDIAN

MAP GRID PROJECTION PLANE

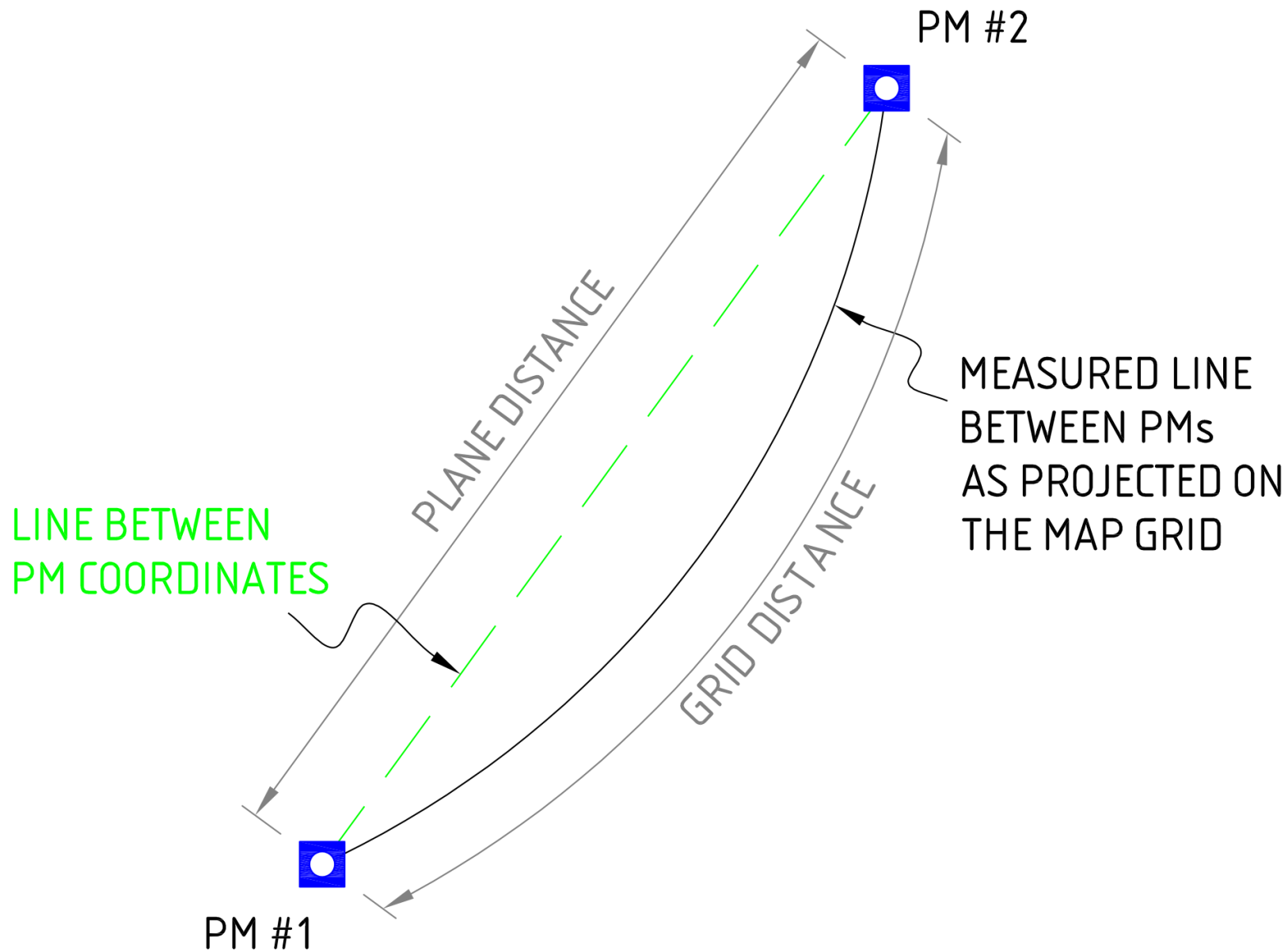


Figure 11: Plane distance and grid distance.

Point Scale Factor / Line Scale Factor

- $LSF = (PSF1 + PSF2) / 2$
- $HSF = \left(1 - \frac{h}{R+h} \right) \quad R \sim 6371 \text{ m @}$

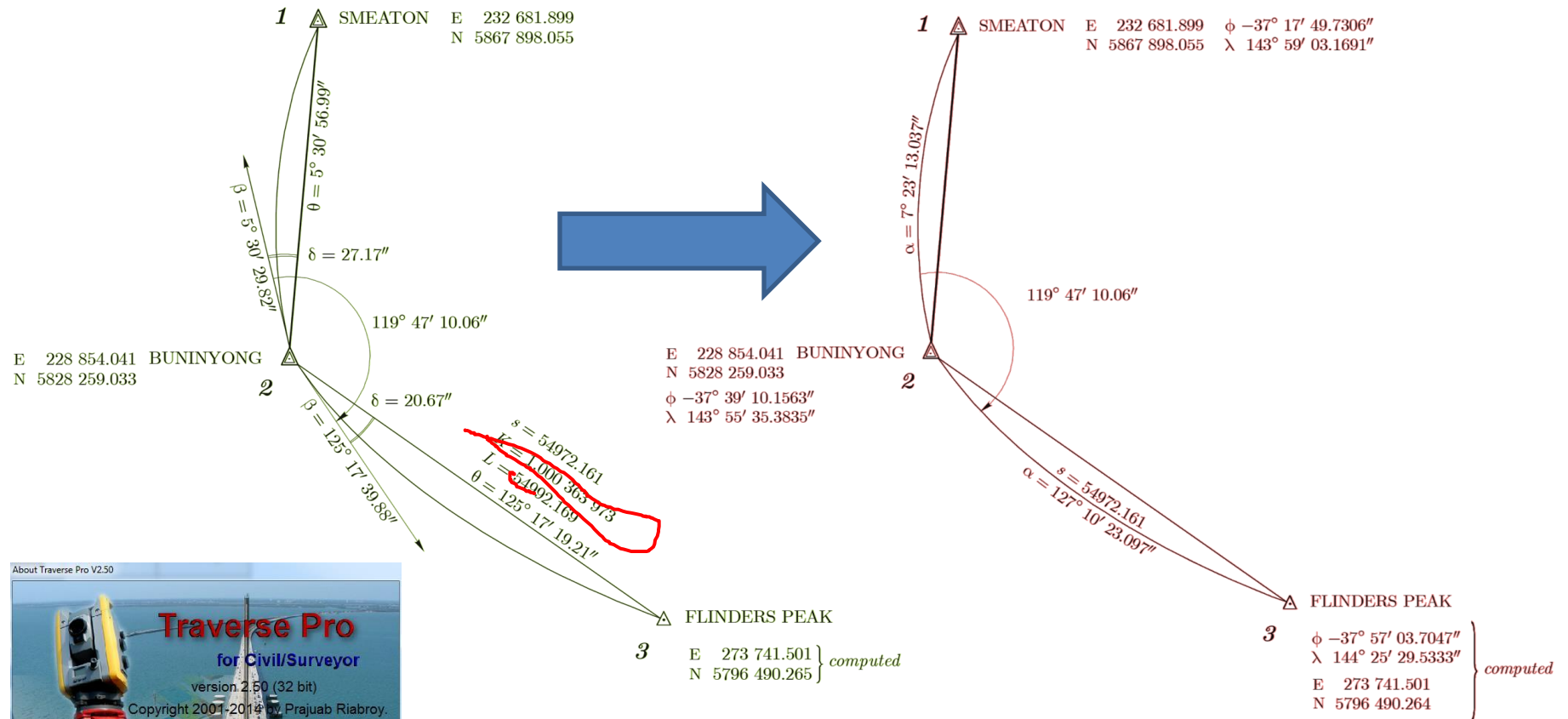
Combined Scale Factor (line) = Height Factor (line) $\times$ line scale factor

- $CSF = HSF * LSF$

GEODETTIC TRAVERSE COMPUTATION

New Approach ! Geodetic Traverse

TRAVERSE COMPUTATION ON THE UTM PLANE



การคำนวณงานวงรอบ 3 แนวทาง

- การคำนวณบนกริดแผนที่ยูทีเอ็ม
 - ยุ่งยาก มีค่าแก้ปริมาณที่รังวัดในสนาม กับปริมาณที่ลดทอนลงบนกริดยูทีเอ็ม
- การคำนวณบนกริด **Low Distortion Projection**
- การคำนวณบนพื้นผิวรูปทรงรีโดยอาศัยเส้นยี่ออเดซิก
- การคำนวณปัญหาทางตรง (**direct problem**)
- การคำนวณปัญหาย้อนกลับ (**inverse problem**)

1 SMEATON E 232 681.899 ϕ $-37^{\circ} 17' 49.7306''$
N 5867 898.055 λ $143^{\circ} 59' 03.1691''$

$\alpha = 7^{\circ} 23' 13.037''$

2 BUNINYONG E 228 854.041 ϕ $-37^{\circ} 39' 10.1563''$
N 5828 259.033 λ $143^{\circ} 55' 35.3835''$

$119^{\circ} 47' 10.06''$

$s = 54972.161$
 $\alpha = 127^{\circ} 10' 23.097''$

3 FLINDERS PEAK ϕ $-37^{\circ} 57' 03.7047''$
 λ $144^{\circ} 25' 29.5333''$
E 273 741.501
N 5796 490.264

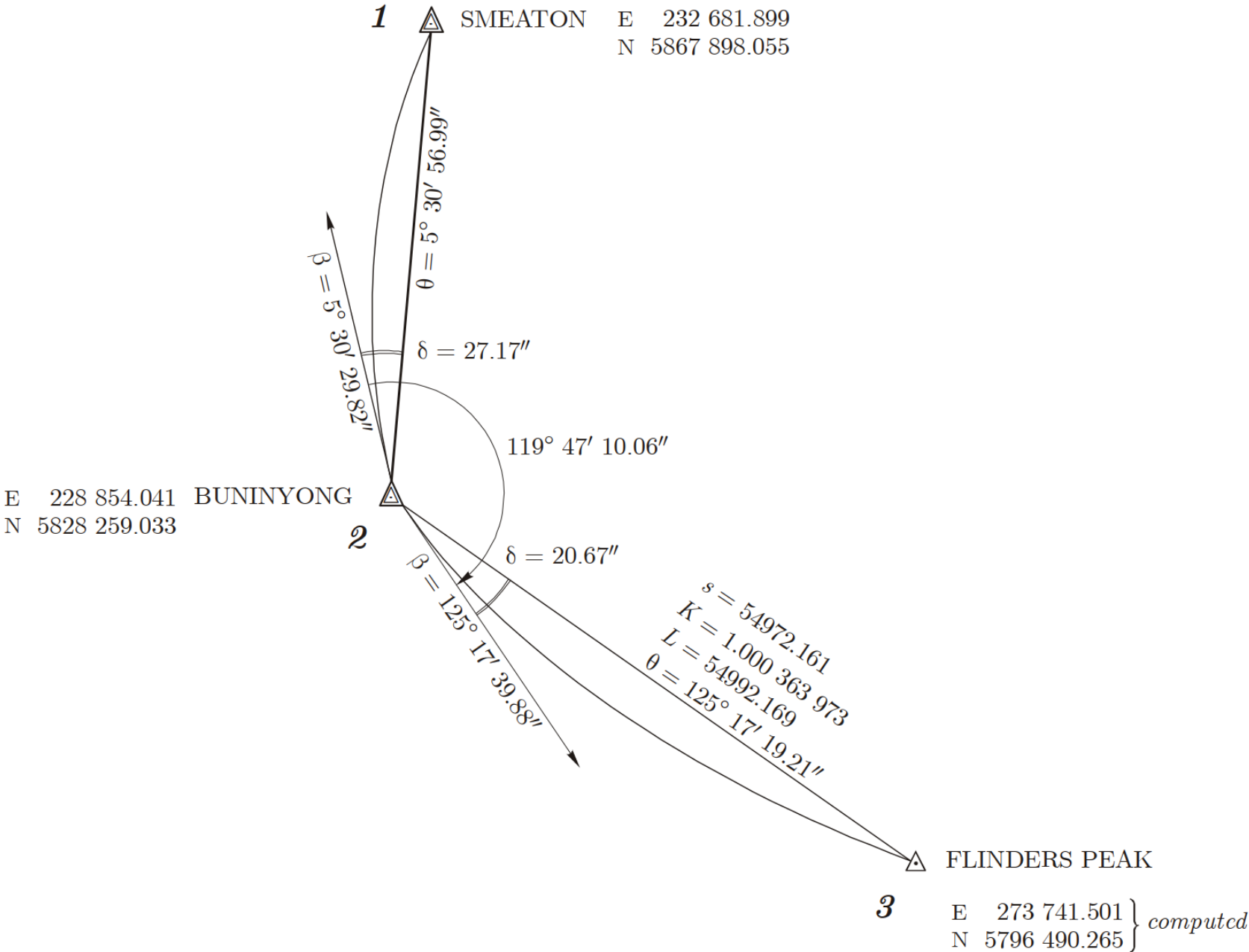
computed

Figure 2. Traverse *Smcaton(1)–Buninyong(2)–Flinders Peak(3)*

การคำนวณวงรอบยี่ห้อเคตีกส์

```
4 import pygeodesy.dms as dms
5 from geographiclib.geodesic import Geodesic
6
7 #####
8 def dd2DMS( dd ):
9     return dms.toDMS( dd, prec=5 )
10
11 def DMS2dd( dms_str ):
12     return dms.parseDMS( dms_str )
13
14
15 geod = Geodesic.WGS84
16 #####
17 SM = DMS2dd("37°17'49.7306S"),DMS2dd("143°59'03.1691E")
18 print( 'SM : ', dd2DMS( SM[0] ) , dd2DMS( SM[1] ) )
19 BY = DMS2dd("37°39'10.1563S"),DMS2dd("143°55'35.3835E")
20 print( 'BY : ', dd2DMS( BY[0] ) , dd2DMS( BY[1] ) )
21 #####
22
23 g21 = geod.Inverse( BY[0], BY[1], SM[0], SM[1] )
24 print( g21 )
25
26 #####
27 Ang_2 = DMS2dd("119°47'10.06")
28 az_23 = g21['azi1'] + Ang_2
29 s_23 = 54_972.161 # angle at #2
30 print('Azimuth BY to FP : ', dd2DMS( az_23 ) )
31 print('Distance BY to FP : ', s_23 )
32
33 #####
34 g_23 = geod.Direct( BY[0], BY[1], az_23, s_23 )
35 print( 'To Flinders Peak: ', g_23 )
36 print( dd2DMS( g_23['lat2'] ) , dd2DMS( g_23['lon2'] ) )
```

TRAVERSE COMPUTATION ON THE UTM PLANE



- (1) Compute the *plane bearing* θ_{21} of the back-sight *Buninyong(2)–Smeaton(1)*.
This is a simple plane coordinate calculation using the known MGA94 coordinates.
 $\theta_{21} = 5^\circ 30' 56.99''$
- (2) Compute the *arc-to-chord* correction δ_{21} of the back-sight *Buninyong(2)–Smeaton(1)*.
The calculation of δ_{21} requires the mean radius r_m , a function of the mean latitude ϕ_m of the back-sight line. ϕ_m can be calculated using **Redfearn.xls** to transform $E, N \Rightarrow \phi, \lambda$ for both ends of the line, taking the mean and then computing r_m followed by
 $\delta_{21} = 27.17''$.
- (3) Compute the *grid bearing* β_{21} of the back-sight *Buninyong(2)–Smeaton(1)*.
 $\beta_{21} = \theta_{21} - \delta_{21} = 5^\circ 30' 29.82''$
- (4) Compute the *grid bearing* β_{23} of the forward-sight *Buninyong(2)–Flinders Peak(3)*.
 $\beta_{23} = \beta_{21} + \psi_2 = 125^\circ 17' 39.88''$ [Note that β_{23} is now known exactly]

Approximate..... And Repeat...

- (5) Compute the E, N coordinates of *Flinders Peak*(3)

This step requires iteration.

- (5.1) The E, N coordinates of *Flinders Peak* are computed using plane trigonometry and the *plane bearing* θ_{23} and *plane distance* L_{23} are required. They are unknown, but can be approximated by $\theta_{23} \simeq \beta_{23}$ and $L_{23} \simeq k_2 s_{23}$ where k_2 is the *point scale factor* at *Buninyong*, a quantity that can be calculated by using **Redfearn.xls** to transform $E, N \Rightarrow \phi, \lambda$ and k, γ for *Buninyong* (γ is the grid convergence). Using these approximations and plane trigonometry, approximate coordinates of *Flinders Peak* can be calculated.

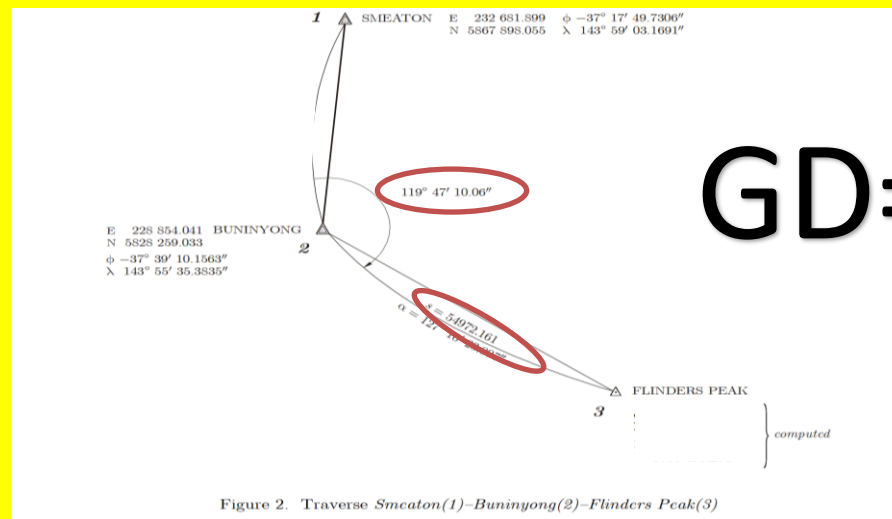
- (5.2) Use **Redfearn.xls** to transform $E, N \Rightarrow \phi, \lambda$ for *Flinders Peak*, then compute ϕ_m and r_m for the line *Buninyong*(2)–*Flinders Peak*(3); then compute the *arc-to-chord correction* δ_{23} and *line scale factor* K_{23} . These will be approximate values (since E, N of *Flinders Peak* and r_m of the line are approximations) but will give better estimates of $\theta_{23} = \beta_{23} + \delta_{23}$ and $L_{23} = K_{23} s_{23}$.

- (5.3) Compute "improved" coordinates of *Flinders Peak* using plane trigonometry and θ_{23}, L_{23} from step (5.2).

Repeat steps (5.2) and (5.3) until there is no change in the coordinates of *Flinders Peak*.

CW-7-A เปรียบเทียบการคำนวณวงรอบยี่ห้อเดทิกส์

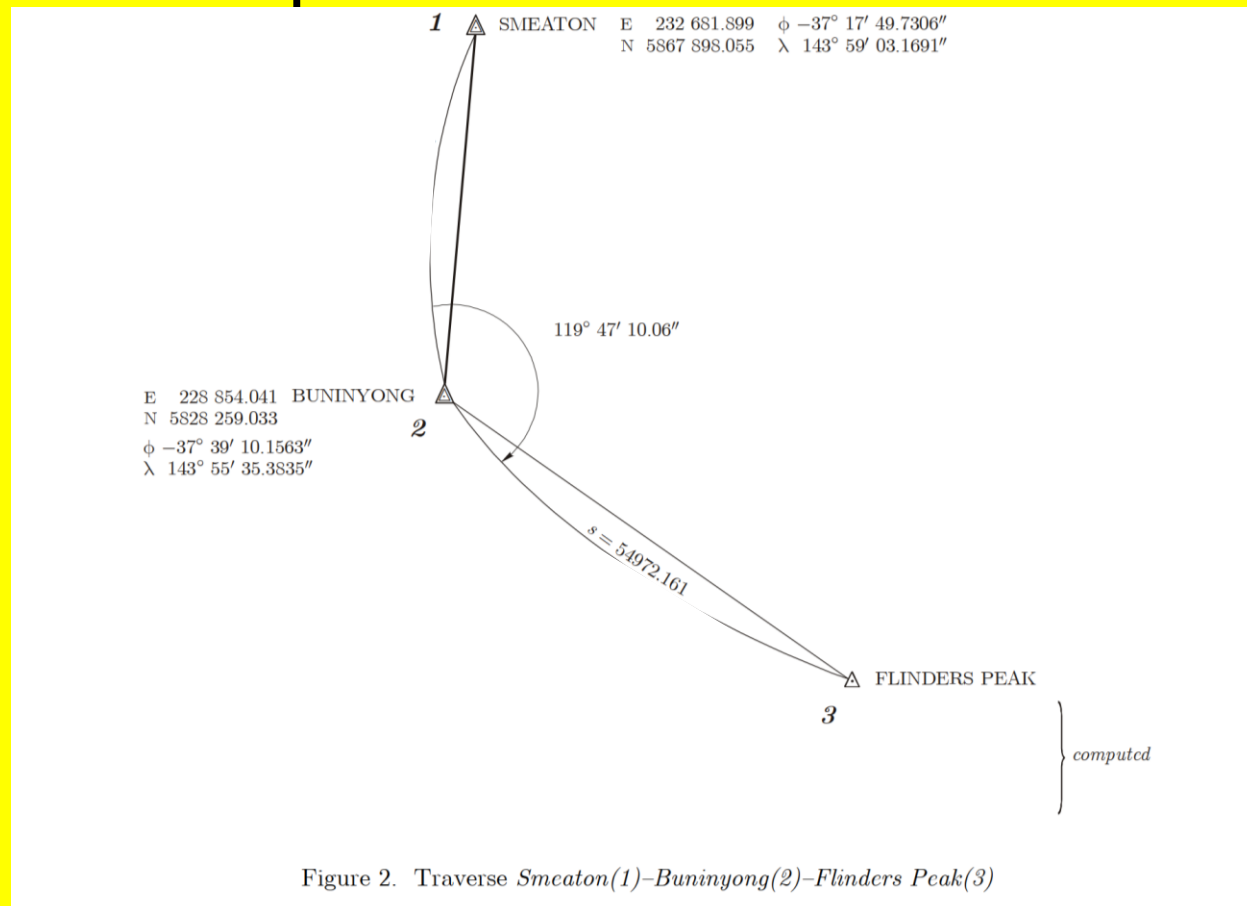
- จงเปรียบเทียบการคำนวณค่าพิกัด **Flinder Peak** โดยคำนวณออกจากหมุดทราบค่าพิกัด **Smeaton** และ **Bunninyong**
- ก) รอบแรกให้คำนวณด้วยระบบพิกัดฉากยูทีเอ็ม ปริมาณนำเข้าในวงสีแดง ให้ศึกษาจากซอฟต์แวร์ “**Geod_UTMTranverse.py**” **scale factor** หากคำนวณจากเฉลี่ย “หัว-ท้าย” จะได้ผลอย่างไร



$$GD = SF * TD$$

CW-7-A เปรียบเทียบการคำนวณวงรอบยี่ห้อเดทิกส์

- ข) คำนวณพิกัดของ Flinder Peak ด้วย direct/inverse problem บน ellipsoid

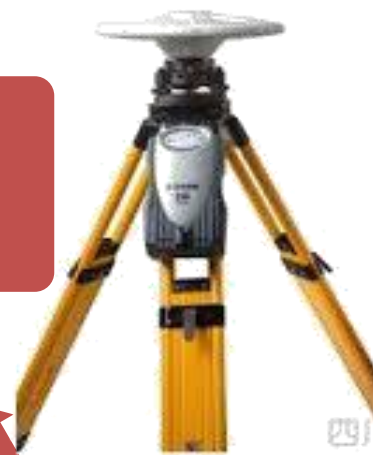


LOW DISTORTION MAP PROJECTION

ระบบพีคัดในการดำเนินงานโครงการ



ECEF($\overline{X}, \overline{Y}, \overline{Z}$)
Geodetic (φ, λ, h)



四川拓泰伟业
www.scttwy.com



UTM (Easting, Northing, H)
TrueDist->Obj.Size(LWH)



LDP (Easting, Northing, H)
No.TrueDist, No SF./
No.t_T



LDP

- การฉายแผนที่ที่ทำให้การคำนวณหน้างาน น้อยลงมากที่สุด
- การทำแผนที่ รังวัดก่อสร้าง ปฏิบัติเสมือนใช้ระบบพิกัดท้องถิ่น
- แต่ยังสามารถโยงยึดกับระบบพิกัดโลก ใช้ประโยชน์จาก **satellite surveying** ได้

เป้าหมาย สเกลแฟกเตอร์ (Scale Factor : SF)

$$\text{Combined SF} = \text{PointSF} * \text{HeightSF}$$



• UTM

- PSF = 0.9996....1.0004
- HSF → มาก ต้องพิจารณาภูมิประเทศ
ที่ตั้ง เทียบกับพื้นผิวทรงรีอ้างอิง (h
ellip.height)

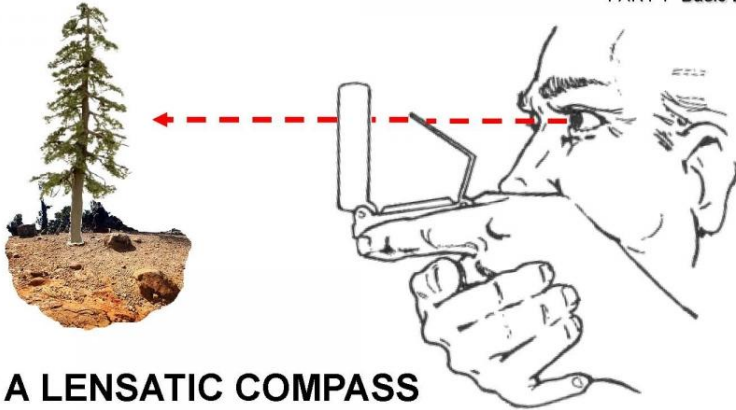
• LDP

- CSF = 0.999 998....1.000 002
- HSF → น้อยมาก เฉลี่ยหาภูมิประเทศ
ที่ตั้งไปบนระนาบแผนที่
- สัดส่วน 1:50,000
- สัดส่วน 1:2,500 **1ppm ... 1 millimeter in 1 kilometer**
- part-per-million < **+/-400 ppm**
- part-per-million < **+/-20 ppm**
(+/-100 ppm)

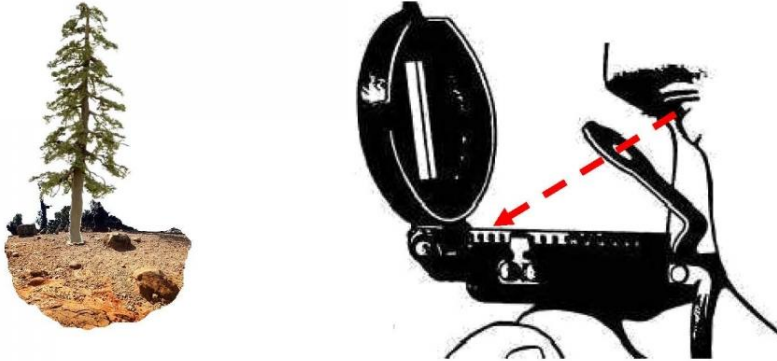
Direction and Orientation

Magnetic Compass

PART 1 Basic Land Navigation

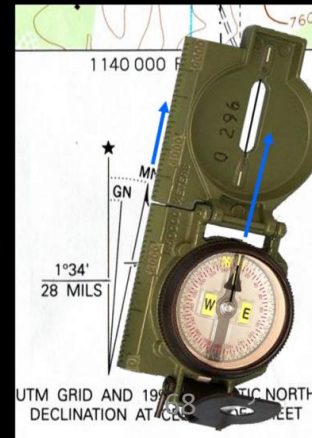
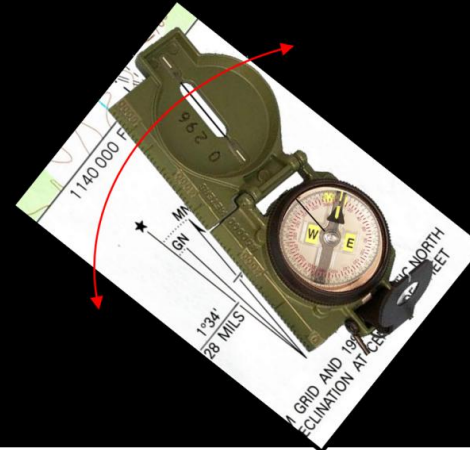


SIGHTING A LENSATIC COMPASS



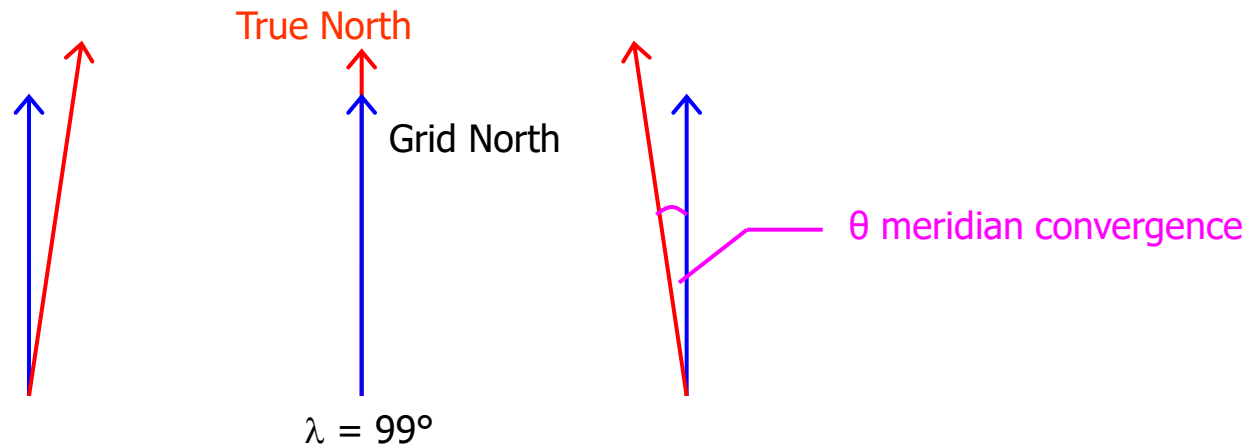
Step One – Orient the map

1. Lay the map on a flat surface and lay the compass on the MN line on the map.
2. **Rotate** map and compass together until the compass bearing reads 0° degrees Magnetic North (compass needle and MN line on the map are **aligned / parallel**). Put rocks on each corner of the map to prevent it from moving again.
3. The map is oriented. The map and its features should appear in the same position that they do in the real world.

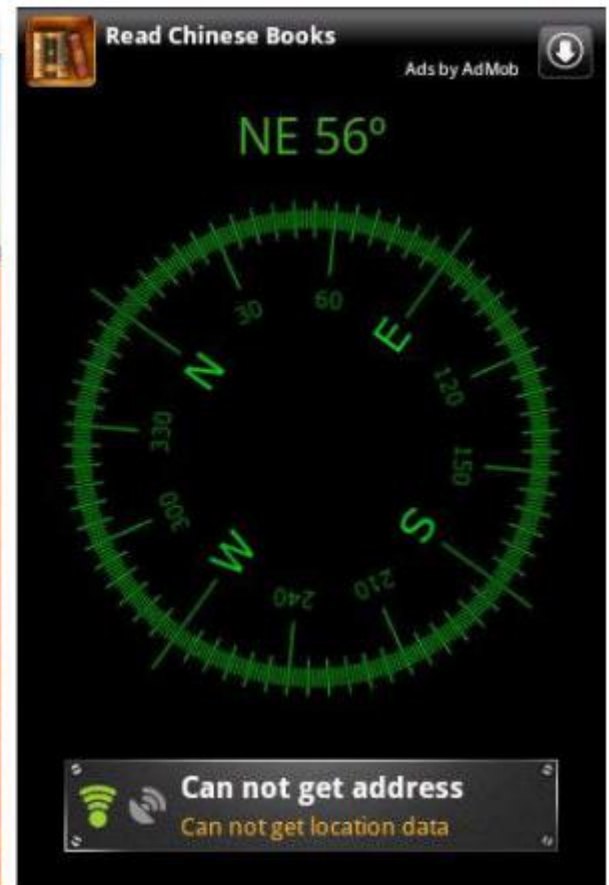
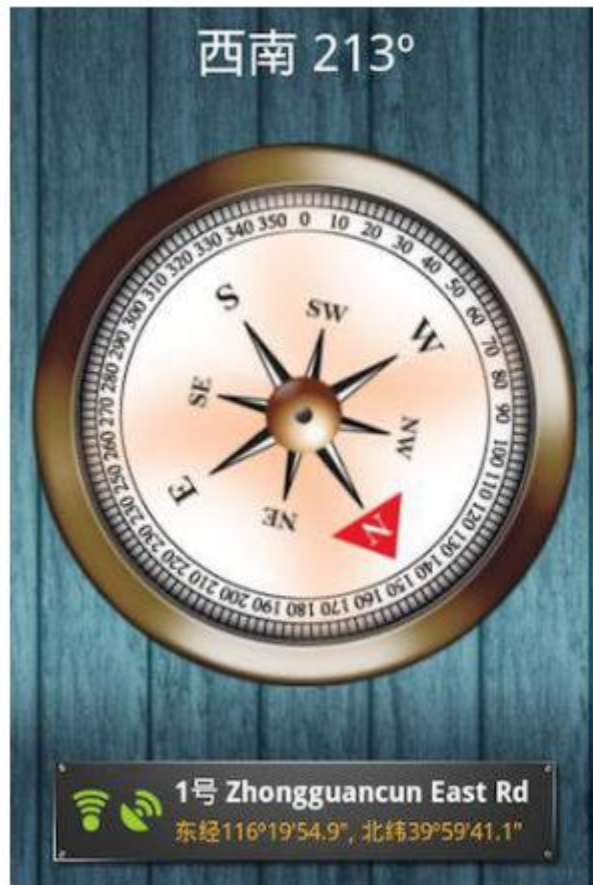


North

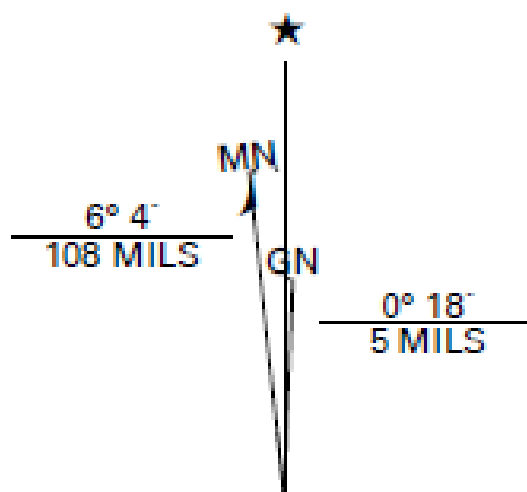
- True North : direction of the North Pole relative to the determined position (MeridianNorth!)
- Grid North : Northward direction of the grid line of the map projection (UTM North)



Electronic Compass



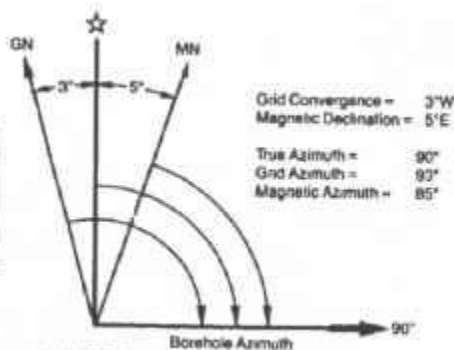




**UTM GRID AND 2012 MAGNETIC NORTH
DECLINATION AT CENTER OF SHEET**

U.S. National Grid
100,000-m Square ID
NJ
Grid Zone Designation
17R

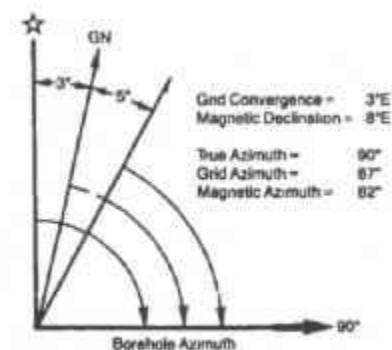
FROM \ TO	True	Grid	Mag
True	0	2	-5
Grid	-2	0	-8
Mag	5	8	0



EXAMPLE 1

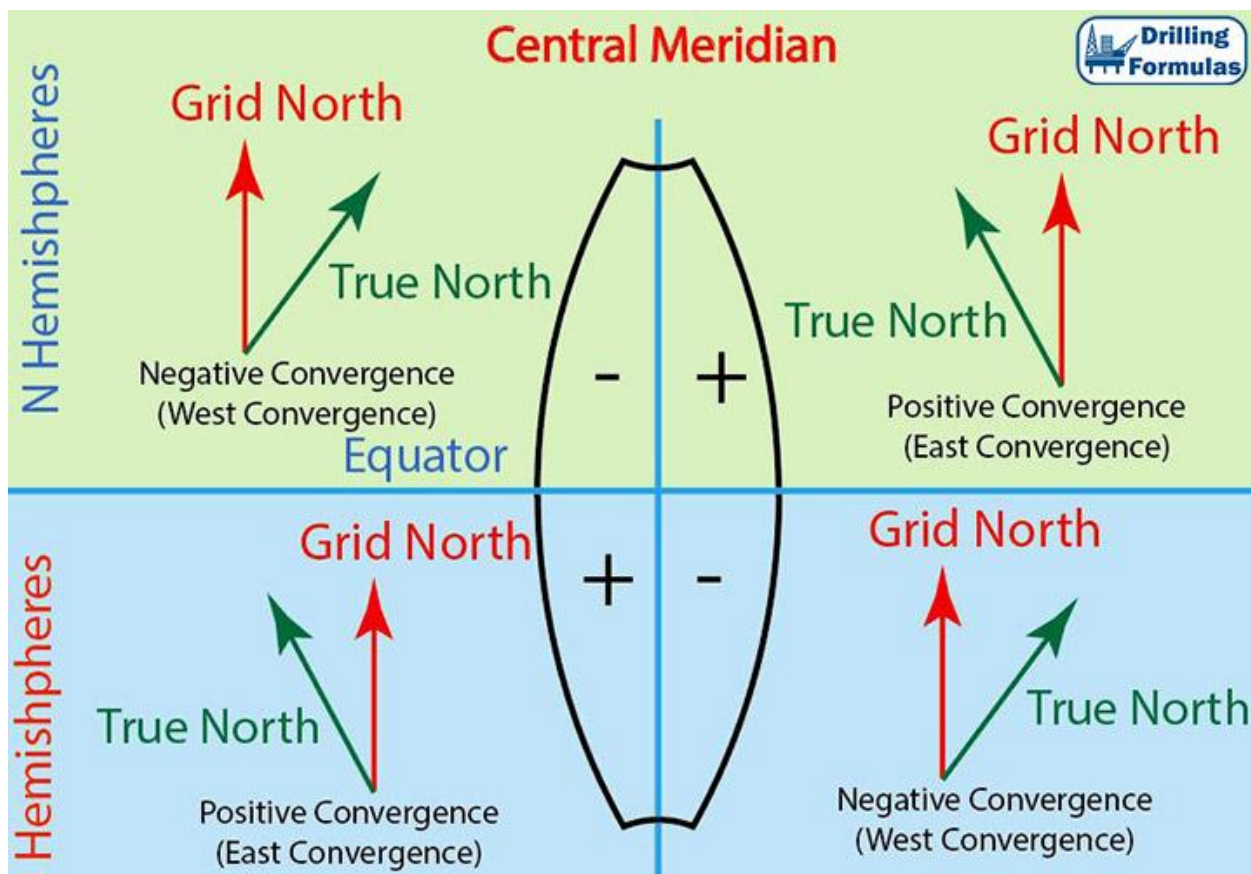
POLAR, GRID/DECLINATION CONVERSION DIAGRAM

FROM \ TO	True	Grid	Mag
True	0	-3	-8
Grid	3	0	-5
Mag	8	5	0

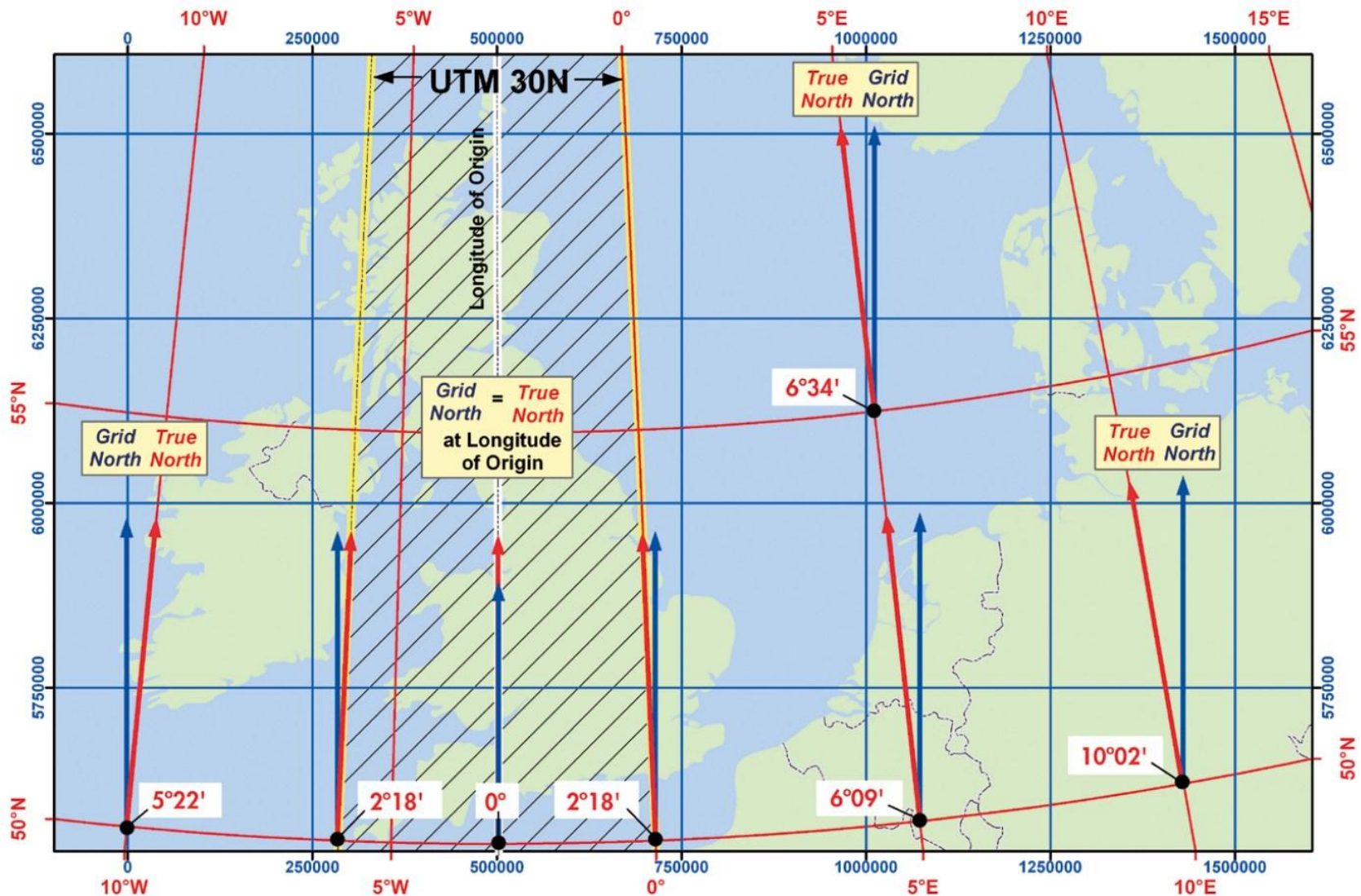


EXAMPLE 2

POLAR, GRID/DECLINATION CONVERSION DIAGRAM



Magnitude of Grid Convergence TM ETRS89/UTM Zone 30N long.origin=3W



Gauss-Bomford convention:

$$\beta = \alpha - \gamma$$

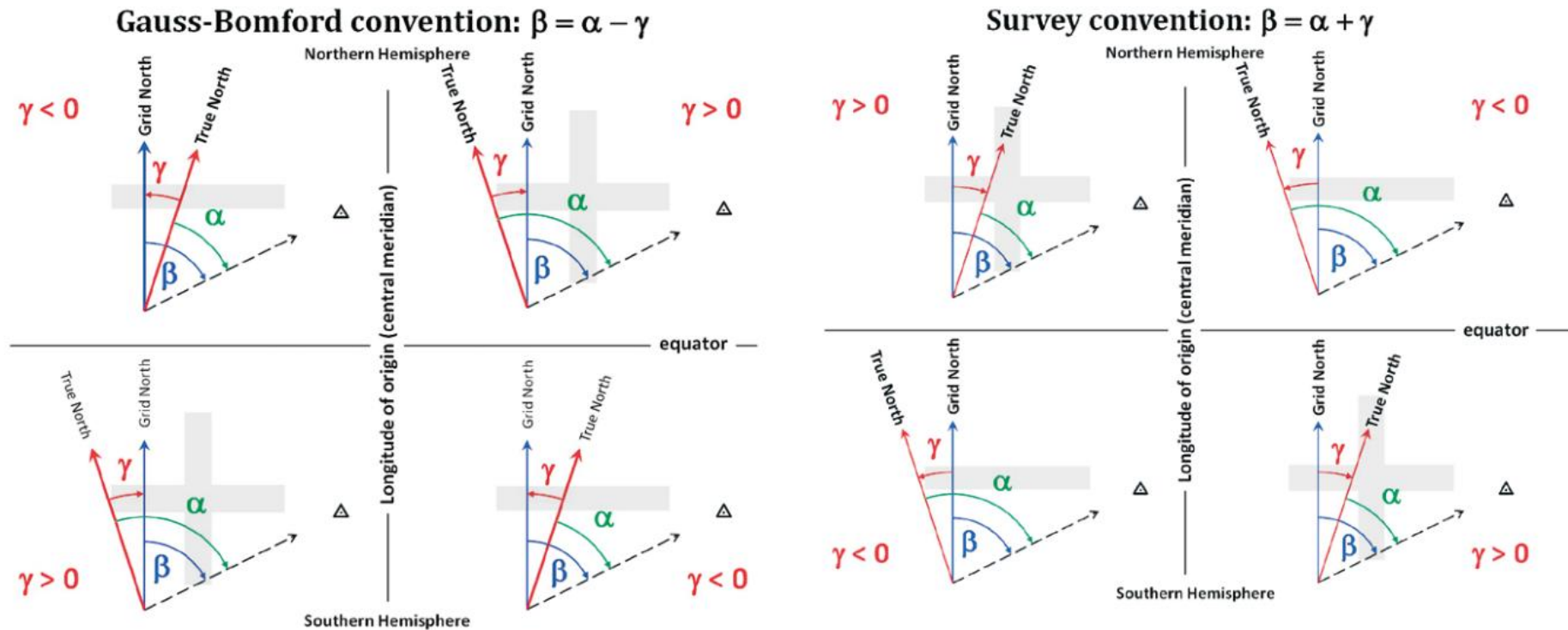
Subtract grid convergence (γ) from azimuth (α) to obtain bearing (β).

Survey convention:

$$\beta = \alpha + \gamma$$

Add grid convergence (γ) to azimuth (α) to obtain bearing (β).

Figure 1: *The sign of grid convergence for transverse cylindrical projections*



The sign of the grid convergence value is shown in each quadrant in grey.

Gazza's Geodetics Calculator For Transverse Mercator Projections

Spheroid:	WGS 1984	Projection:	UTM North	False Northing:	0
Semi-Major Axis:	6378137	UTM Zone No.:	47	False Easting:	500000
Inverse Flattening:	298.257223563	Longitude of Central Meridian:	99°	Central Scale Factor:	0.9996
				Grid Units:	Metres

Latitude (- south):	14	0'	0		Lat/Lon to N/E	Northing:	1547954.32
Longitude (- west):	100	0'	0		N/E to Lat/Lon	Easting:	607995.652
Grid Convergence:	0°	-14'	31.0037	Point 1		Point Scale Factor:	0.99974426

Ellipsoidal Distance:	0		Grid Distance:	0
True Bearing:	0°	0'	Grid Bearing:	0°
		Calc R & B	Calc Point 2	
			Calc R & B	
			Calc Point 2	

Latitude (- south):	0°	0'	0		Lat/Lon to N/E	Northing:	0
Longitude (- west):	0°	0'	0		N/E to Lat/Lon	Easting:	0
Grid Convergence:	0°	0'	0	Point 2		Point Scale Factor:	0
				Swap 1 & 2			
				Reset Fields		Swap 1 & 2	

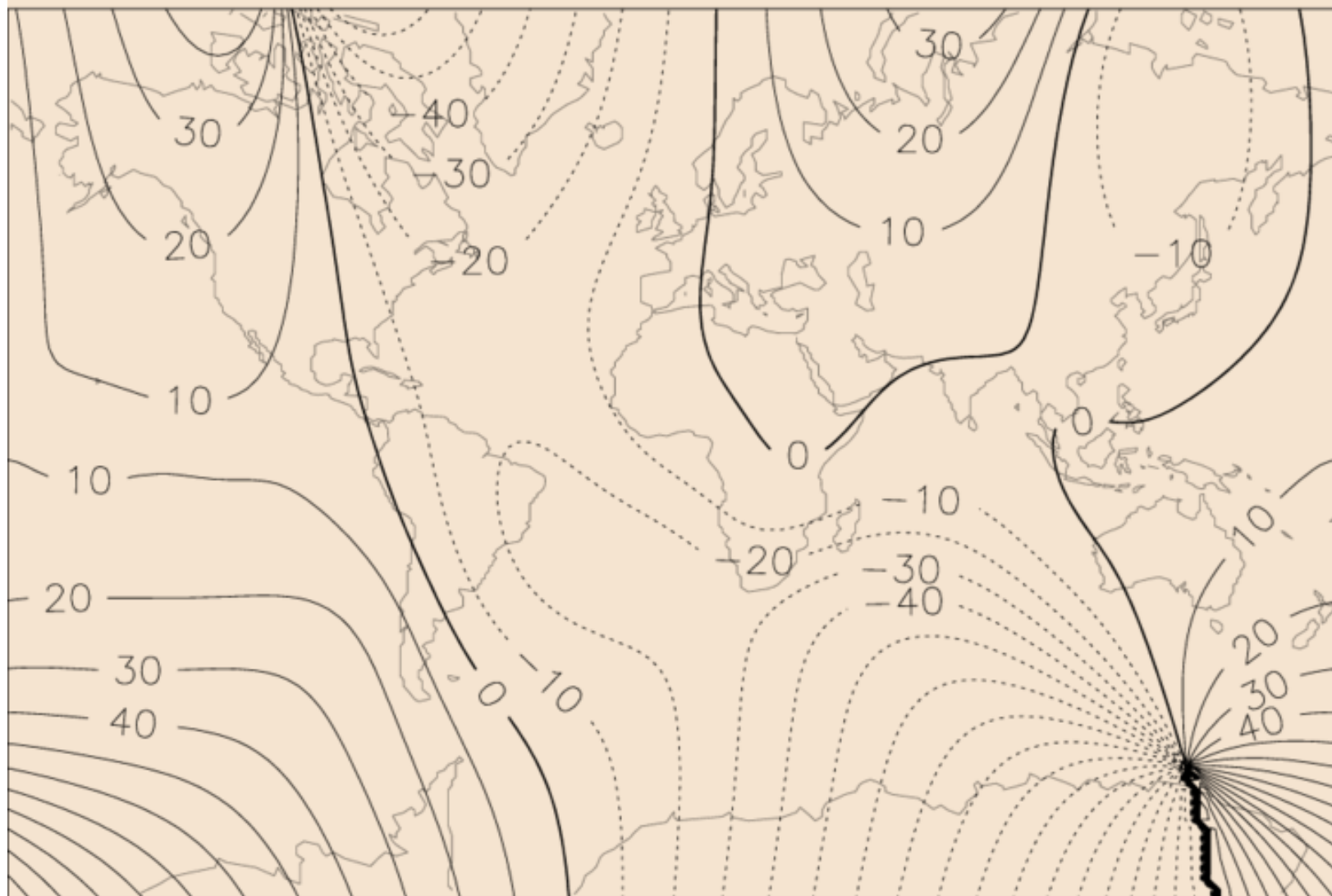
NORTH DIRECTION



Magnetic North

- Direction to the North according to the Earth's magnetic field
- Several errors may induced 'declination' and local attraction.
- Achieved via metal compass or electronic compass , accuracy ranging from +/- 5 degree to 0.5 degree

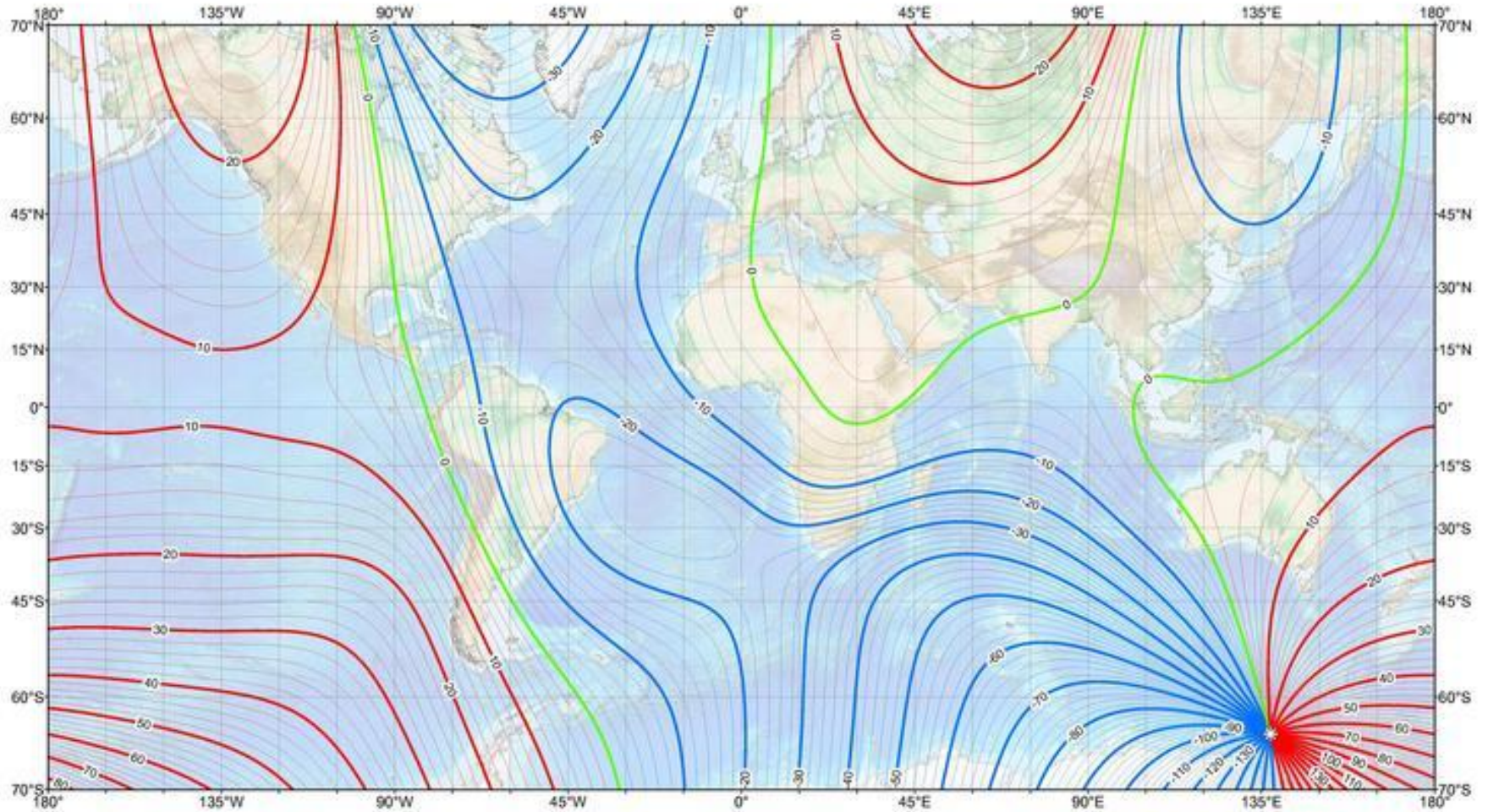
2000 Declination (degrees east)



<http://geomag.usgs.gov>

International Geomagnetic Reference Field (IGRF)

US/UK World Magnetic Model -- Epoch 2010.0 Main Field Declination (D)



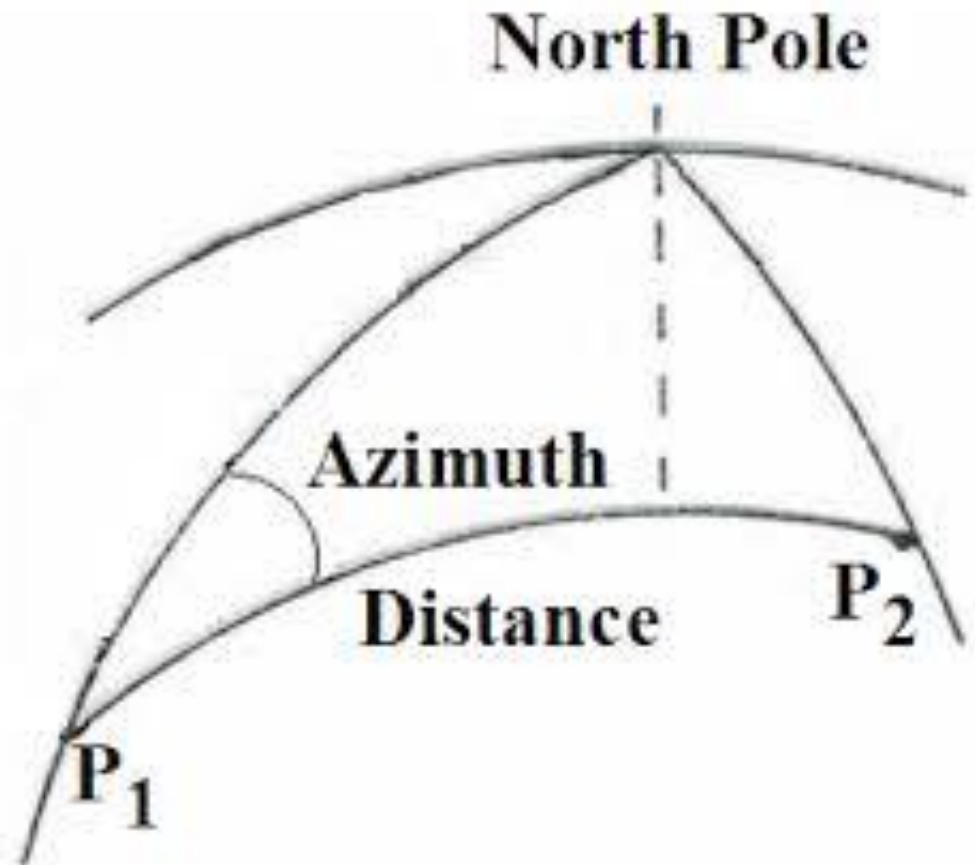
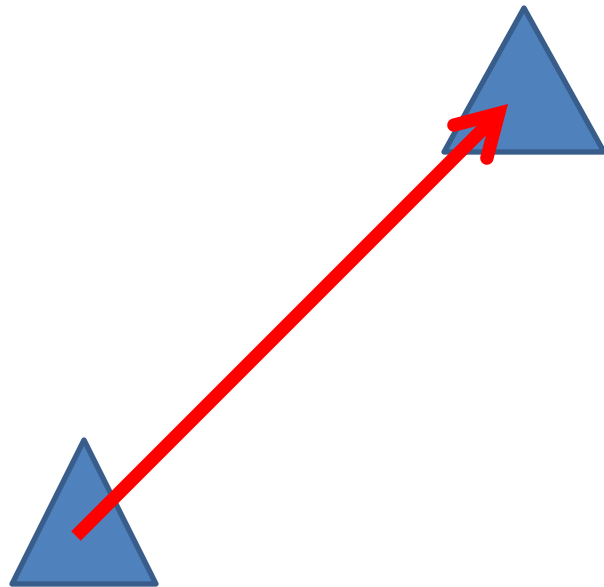
Main field declination (D)
Contour interval 2 degrees, red contours positive (east); blue negative (west); green (agonic) zero line.
Mercator Projection.
☉ : Position of dip poles

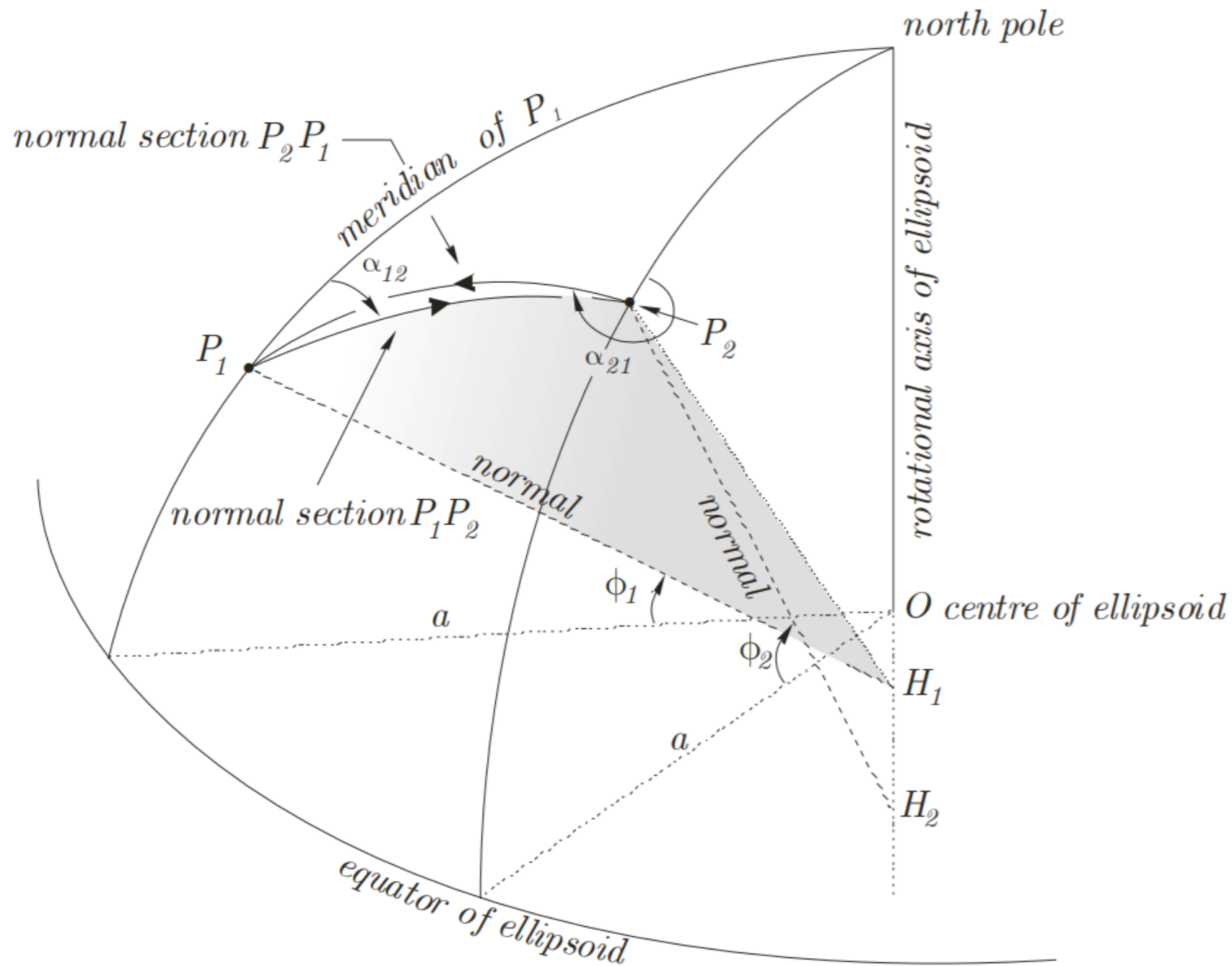
Map developed by NOAA/NGDC & CIRES
<http://ngdc.noaa.gov/geomag/WMM/>
Map reviewed by NGA/BGS
Published January 2010

'GPS' North

- Picking two coordinates from GPS and calculate direction results 'True North' or azimuth.
- Thus GPS North == True North !
- Since handheld GPS is +/- 10 meter accuracy, the two points has to be adequate distanced.
- Geodetic Fundamental 'Direct' Problem

Grid Azimuth vs. True Azimuth





ClassWork 6-6 True and Grid Azimuth

- ให้นิสิตเลือกสนามบินตามหมายเลข CR58 ตรวจสอบอักษรย่อ 3 ตัว ค้นหาด้วย '@@@ airport coordinate' คนละหนึ่งแห่ง เปิด google maps แล้วเลือก "ทางวิ่งหลัก" ที่ยาวที่สุด 1 แห่ง
- ทำการสเกตส์ ขนาดและทิศทางวิ่ง แล้วอ่านพิกัด ต้นทาง @@-L ไปยัง ปลายทาง @@-R ด้วย 'What's here ?' เช่น 43.674403, -79.663716 และ 43.694214, -79.634931
- นำค่าพิกัด @@-L ไปยังปลายทาง @@-R คำนวณแอดมิชชัน อ้างอิงจาก local meridian (α_{12}) ด้วย inverse problem คำสั่ง geograhiclib/geodsolve คำนวณละเอียดถึง 1 arcsecond
- ให้นิสิตคำนวณพิกัดทั้งสอง @@-L ไปยังปลายทาง @@-R แปลงไปเป็น Zone/UTM geograhiclib/geoconvert แล้วคำนวณทิศทางด้วยพิกัดกริดยูทีเอ็ม แอดมิชชันนี้เรียกว่า กริดแอดมิชชัน Az12
- จากนั้นให้นิสิตคำนวณ meridian convergence (γ) จากพิกัดทั้งสอง @@-L ไปยังปลายทาง @@-R ด้วย geograhiclib/geoconvert -c -p X พิจารณา หน่วย และความละเอียดที่ได้ บันทึกค่า
- หาความสัมพันธ์ของ true azimuth และ grid azimuth และ meridian convergence แล้วให้คำนวณผลต่างกับ คำสั่ง geograhiclib/geodsolve คำนวณละเอียดถึง 1 arcsecond

Azimuth Accuracy

- E.g. GPS Handheld +/- 10m

	10	1	
Distance	Azim.Error (m)		
100	5.73	0.57	
200	2.86	0.29	
300	1.91	0.19	
400	1.43	0.14	
500	1.15	0.11	
600	0.95	0.10	
700	0.82	0.08	
800	0.72	0.07	

IGRF

- The International Geomagnetic Reference Field (IGRF) is a standard mathematical description of the Earth's main magnetic field and its secular variation. It is the product of a collaborative effort between magnetic field modellers and the institutes involved in collecting and disseminating magnetic field data from satellites and from observatories and surveys around the world.

The World Magnetic Model (WMM)

- The World Magnetic Model is the standard model used by the U.S. DoD, the U.K. Ministry of Defence, the NATO and IHO, for navigation, attitude and heading referencing systems using the geomagnetic field.
- The model, associated software, and documentation are distributed by NCEI on behalf of NGA.
- The model is produced at 5-year intervals, with the current model expiring on December 31, 2019. → WMM-2015

Geographiclib/MagneticField

- **MagneticField** reads in times and positions on standard input and prints out the geomagnetic field on standard output and, optionally, its rate of change.
- The magnetic field from WMM2015 in Timbuktu on 2016-01-01

```
GeographicLib-1.47\bin>MagneticField -r --  
input-string "2017-08-29 14:00:00N 100:00:00E 0"  
-0.59 16.05 41022.9 41020.8 -419.0 11800.3  
42686.4 0.01 0.20 -4.8 -4.8 6.9 150.6 37.0
```

NOAA/Magnetic Field Calculators

Magnetic Declination Estimated Value

Declination is calculated using the most recent [World Magnetic Model \(WMM\)](#) or the [International Geomagnetic Reference Field \(IGRF\)](#). Prior to 1900 the calculator is based on the [gufm1](#) model. A smooth transition from gufm1 to IGRF is available from 1900 to 1999, but accurate to 30 minutes of arc, but environmental factors can cause magnetic field distortion. The calculator can output results in HTML, XML, or CSV programmatically (API). For more information click the information icon.

Calculate Declination

Latitude: ☐ S ☒ N

Longitude: ☐ W ☒ E

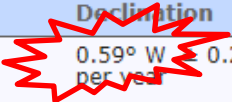
Model: ☒ WMM (2014-2019) ☐ IGRF (1590-2019)

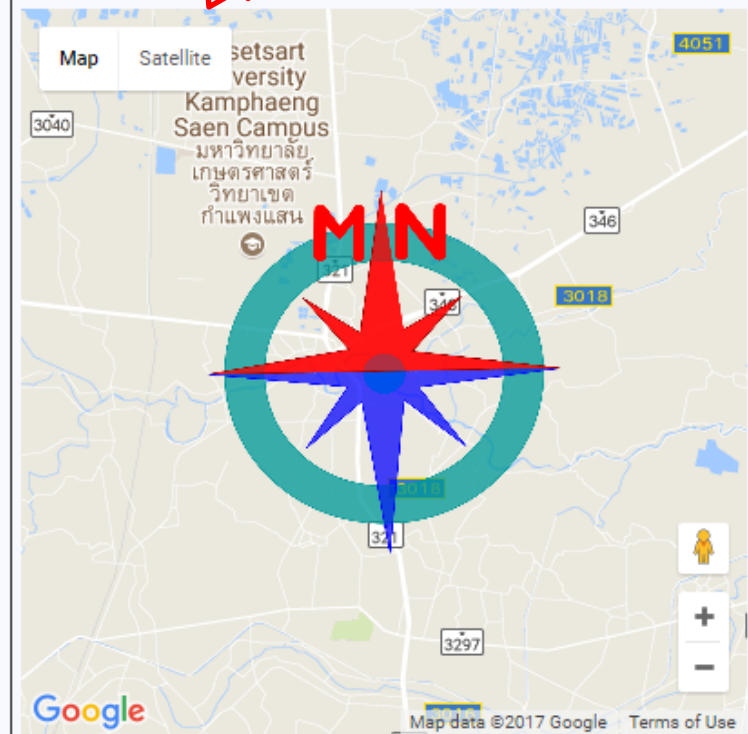
Date: Year Month Day

Result format: ☒ HTML ☐ XML ☐ CSV ☐ PDF

Calculate

Declination

Model Used:	WMM2015
Latitude:	14° N 
Longitude:	100° E
Date	Declination
2017-08-29	0.59° W  0.27° changing by 0.01° E per year



NGDC Declination

Date 2014-11-03

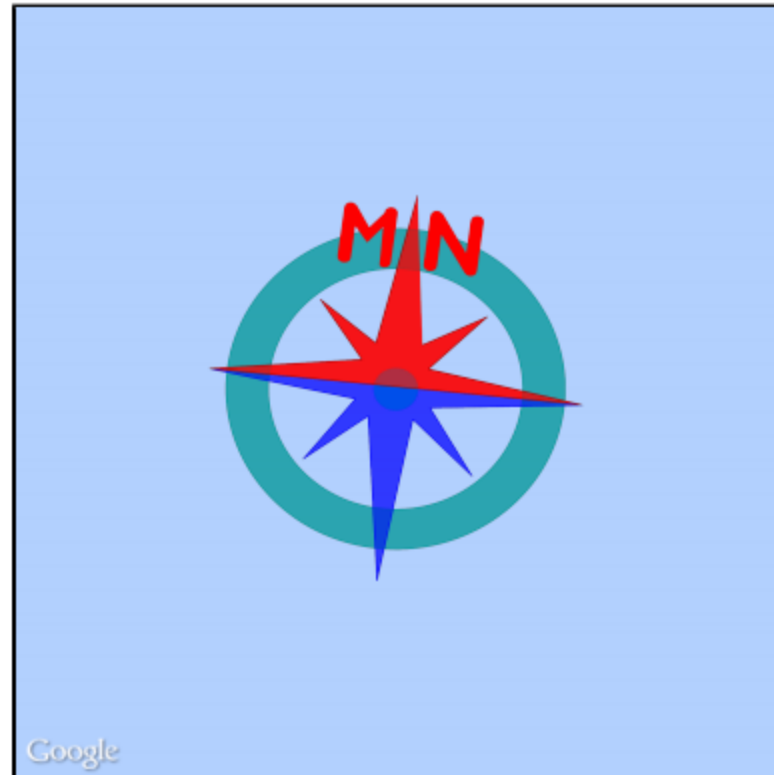
Latitude 14° N

Longitude 100° W

Elevation 0.0 km

Model Used WMM 2010

Declination 5.62° E changing by
0.1° W per year



Compass shows the approximate bearing of the
magnetic north (MN)

NGDC Declination

Date 2014-11-03

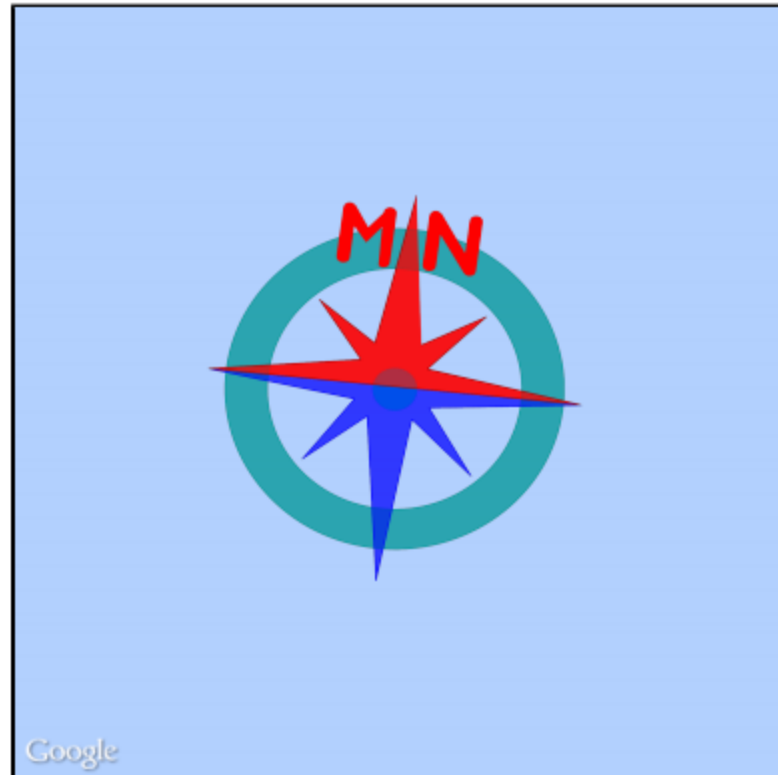
Latitude 14° N

Longitude 100° W

Elevation 0.0 km

Model Used IGRF 11

Declination 5.57° E changing by
0.11° W per year



Compass shows the approximate bearing of the
magnetic north (MN)

ClassWork : Runway Azimuth



ClassWork 6-7 True Azimuth and Compass Direction

- ให้นิสิตเลือกสนามบินตามหมายเลข CR58 ตรวจสอบอักษรย่อ 3 ตัว ค้นหาด้วย '@@@ airport coordinate' คนละหนึ่งแห่ง เปิด google maps แล้วเลือก "ทางวิ่งหลัก" ที่ยาวที่สุด 1 แห่ง
- ทำการสเกตส์ ขนาดและทิศทางวิ่ง แล้วอ่านพิกัด ต้นทาง @@-L ไปยัง ปลายทาง @@-R ด้วย 'What's here ?' เช่น 43.674403, -79.663716 และ 43.694214, -79.634931
- นำค่าพิกัด @@-L ไปยังปลายทาง @@-R คำนวณแอดิมัธ อ้างอิงจาก true north (α_{12}) ด้วย inverse problem คำสั่ง geograhiclib/geosolve
- ให้นิสิตคำนวณพิกัดทั้งสอง @@-L ไปยังปลายทาง @@-R ไปคำนวณ magnetic declination ทั้งสองจุด geograhiclib/magnetic ให้สังเกตความละเอียดถูกต้องที่ magnetic ให้มา
- หาความสัมพันธ์ของ true azimuth , compass direction และ magnetic declination
- ในอุปกรณ์มือถือ smart phone นิสิตใช้ application ใด ที่ บอก electronic compass และมีวิธีการ calibration อย่างไร

MGRS : MILITARY GRID REFERENCE SYSTEM

MGRS / USNG

- Military Grid Reference System (MGRS)
- United State National Grid (USNG)
- Derived from UTM coordinate

An example of an *MGRS coordinate*, or *grid reference*, would be 4QFJ12345678 , which consists of three parts:

- 4Q (grid zone designator, GZD)
- FJ (the 100,000-meter square identifier)
- 12345678 (numerical location; **easting** is 1234 and **northing** is 5678, in this case specifying a location with 10 m resolution)

MGRS for different precision

- 4QGZD only, precision level $6^{\circ} \times 8^{\circ}$ (in most cases)
- 4QFJGZD and 100 km Grid Square ID, precision level 100 km
- 4QFJ 1 6precision level 10 km
- 4QFJ 12 67precision level 1 km
- 4QFJ 123 678precision level 100 m
- 4QFJ 1234 6789precision level 10 m
- 4QFJ 12345 67890precision level 1 m

GeographicLib / geoconvert

- `>geoconvert -u -p 0 --input-string "14.000
100.000"`

47n 607996 1547954

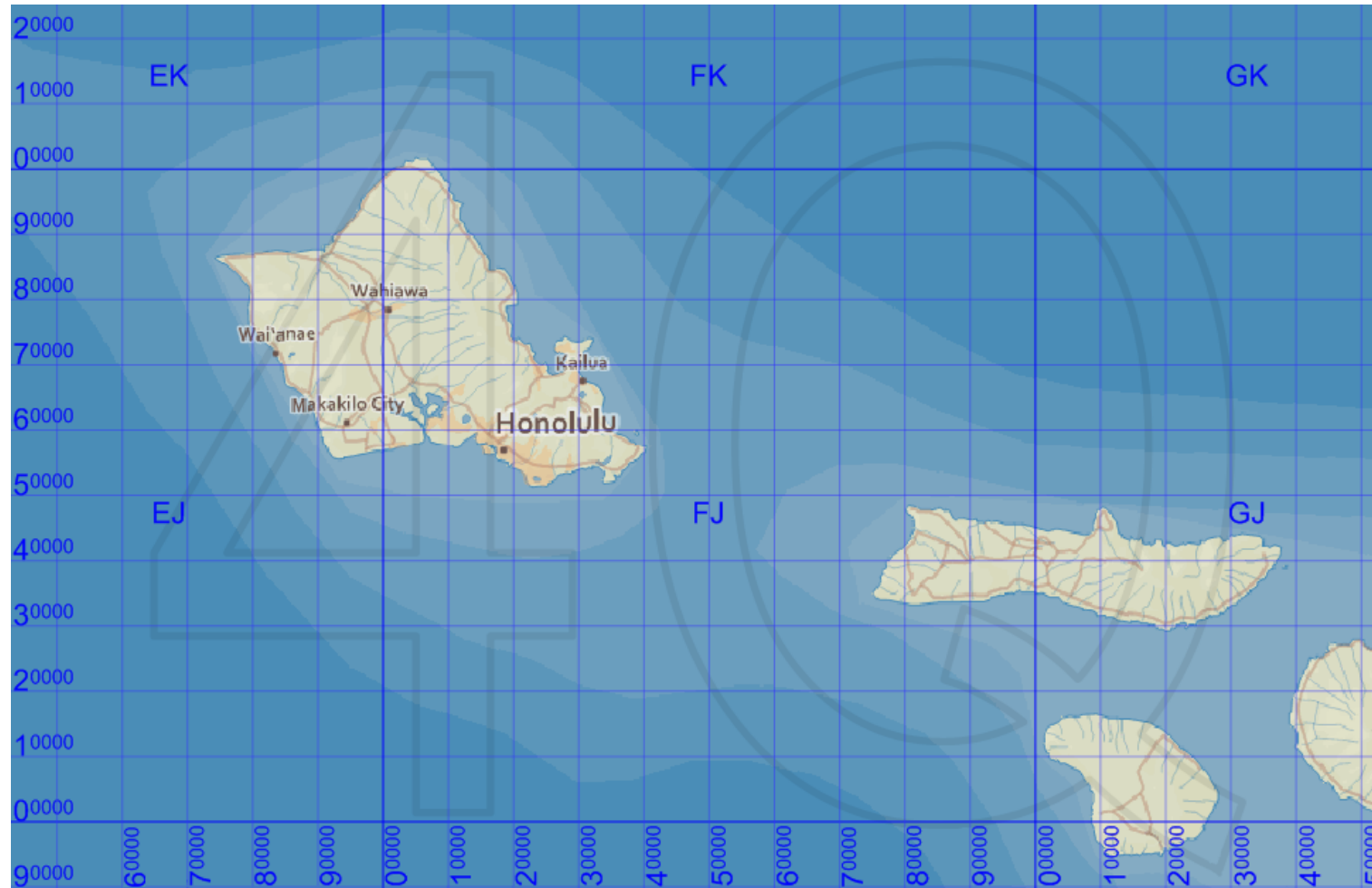
- `>geoconvert -m -p 0 --input-string "14.000
100.000"`

47PPR0799547954

- `>geoconvert -m -p -2 --input-string "14.000
100.000"`

47PPR079479

100 km square that is called **4QFJ15**.



Geohash

- public domain geocoding system invented by Gustavo Niemeyer[1], which encodes a geographic location into a short string of letters and digits.

geohash length	lat bits	lng bits	lat error	lng error	km error
1	2	3	± 23	± 23	± 2500
2	5	5	± 2.8	± 5.6	± 630
3	7	8	± 0.70	± 0.70	± 78
4	10	10	± 0.087	± 0.18	± 20
5	12	13	± 0.022	± 0.022	± 2.4
6	15	15	± 0.0027	± 0.0055	± 0.61
7	17	18	± 0.00068	± 0.00068	± 0.076
8	20	20	± 0.000085	± 0.00017	± 0.019

<https://www.movable-type.co.uk/scripts/geohash.html>

Geohash

Enter latitude, longitude & precision to obtain geohash; enter geohash to obtain latitude/longitude.

Latitude /
Longitude

Precision

5 characters ▾

Geohash

Geohash length *Cell width* *Cell height*

1 ≤ 5,000km × 5,000km

2 ≤ 1,250km × 625km

3 ≤ 156km × 156km

4 ≤ 39.1km × 19.5km

5 ≤ 4.89km × 4.89km

6 ≤ 1.22km × 0.61km

7 ≤ 153m × 153m

8 ≤ 38.2m × 19.1m

9 ≤ 4.77m × 4.77m

10 ≤ 1.19m × 0.596m

11 ≤ 149mm × 149mm

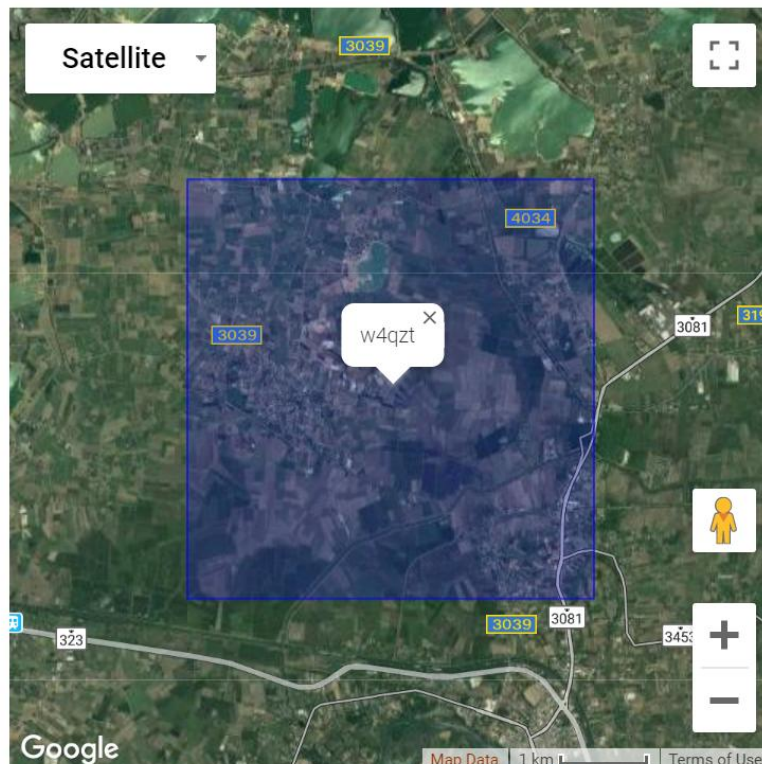
12 ≤ 37.2mm × 18.6mm

Neighbours:

w4qzu w4qzv w4qzy

w4qzs **w4qzt** w4qzw

w4qzk w4qzm w4qzq



Python / Geohash

- **PyGeodesy**
- A pure Python implementation of geodesy tools for various ellipsoidal and spherical earth models using precision trigonometric, vector-based, elliptic and approximate methods for geodetic (lat-/longitude) and geocentric ([ECEF](#) cartesian) coordinates.

*bounds(geohash, LatLon=None, **kws)*

Returns the lower-left SW and upper-right NE corners of a geohash.

Parameters:

- **geohash** - To be bound ([Geohash](#)).
- **LatLon** - Optional (sub-)class to return the bounds (LatLon) or None.
- **kws** - Optional keyword arguments for **LatLon**.

Returns:

A [Bounds2Tuple](#)(latlonSW, latlonNE) as **LatLon** or a [Bounds4Tuple](#)(

Raises:

- **TypeError** - The **geohash** is not a [Geohash](#), LatLon or str.
- [GeohashError](#) - Invalid or null **geohash**.

Example:

```
>>> geohash.bounds('u120fxw') # 52.20428467, 0.11810303,
                                # 52.20565796, 0.11947632
>>> geohash.decode('u120fxw') # '52.205', '0.1188'
```

decode(geohash)

Decode a geohash to lat-/longitude of the (approximate centre of) geohash cell, to reasonable precision.

Parameters:

- **geohash** - To be decoded ([Geohash](#)).

Returns:

2-Tuple "(latStr, lonStr)" in (str).

Raises:

- **TypeError** - The **geohash** is not a [Geohash](#), LatLon or str.
- [GeohashError](#) - Invalid or null **geohash**.

Example:

```
# '52.205', '0.1188'
# '23.708', '42.473' Saudi Arabia
# '69.6', '-45.7' Greenland
# '-24.87', '162.95' Coral Sea
# '65.48', '-17.75' Iceland
```

encode(lat, lon, precision=None)

Encode a lat-/longitude as a geohash, either to the specified precision or if not provided, to an automatically evaluated precision

Parameters:

- **lat** - Latitude (degrees).
- **lon** - Longitude (degrees).
- **precision** - Optional, the desired geohash length (int 1..12).

Returns:

The geohash (str).

Raises:

- [GeohashError](#) - Invalid **lat**, **lon** or **precision**.

Example:

```
>>> geohash.encode(52.205, 0.119, 7) # 'u120fxw'
>>> geohash.encode(52.205, 0.119, 12) # 'u120fxwshvkg'
>>> geohash.encode(52.205, 0.1188, 12) # 'u120fxws0jre'
>>> geohash.encode(52.205, 0.1188) # 'u120fxw'
>>> geohash.encode(0, 0) # 's00000000000'
```

neighbors(geohash)

Return the [Geohashes](#) for all 8 adjacent cells.

Parameters:

- **geohash** - Cell for which neighbors are requested ([Geohash](#) or str).

Returns:

A [Neighbors8Dict](#)(N, NE, E, SE, S, SW, W, NW) of [Geohashes](#).

Raises:

- **TypeError** - The **geohash** is not a [Geohash](#), LatLon or str.

JS name: *neighbours*.

ClassWork 6-3 : UTM MGRS Geohash

A4 UTM

- ให้เขียนเส้น สีแดง แทนเมอริเดียนกลางผ่านไทย
- ให้คำนวณพิกัดยูทีเอ็มละเอียดถึง “เมตร”
- สำหรับกริดพิกัด “องศา” ในพื้นที่ประเทศไทย

A4 MGRS

- ให้คำนวณพิกัด **MGRS** ละเอียดถึง **100** เมตร
- และให้คำนวณสเกลแฟกเตอร์ทุกจุด

